Computational Engineering

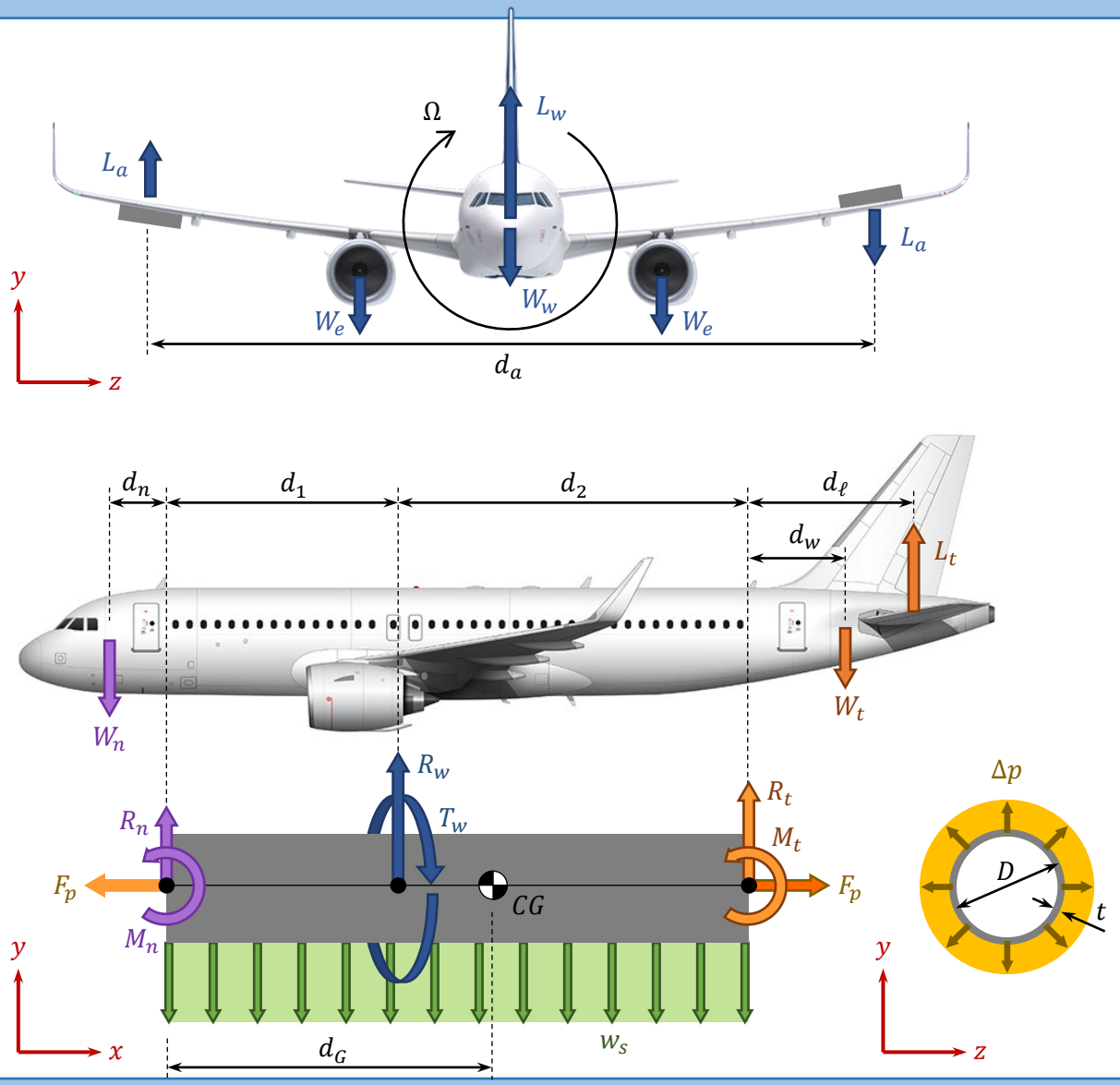
Computational Solid Mechanics (CSM)

Juan Carlos Cante, David Roca, and Daniel Yago

Motivation and objectives

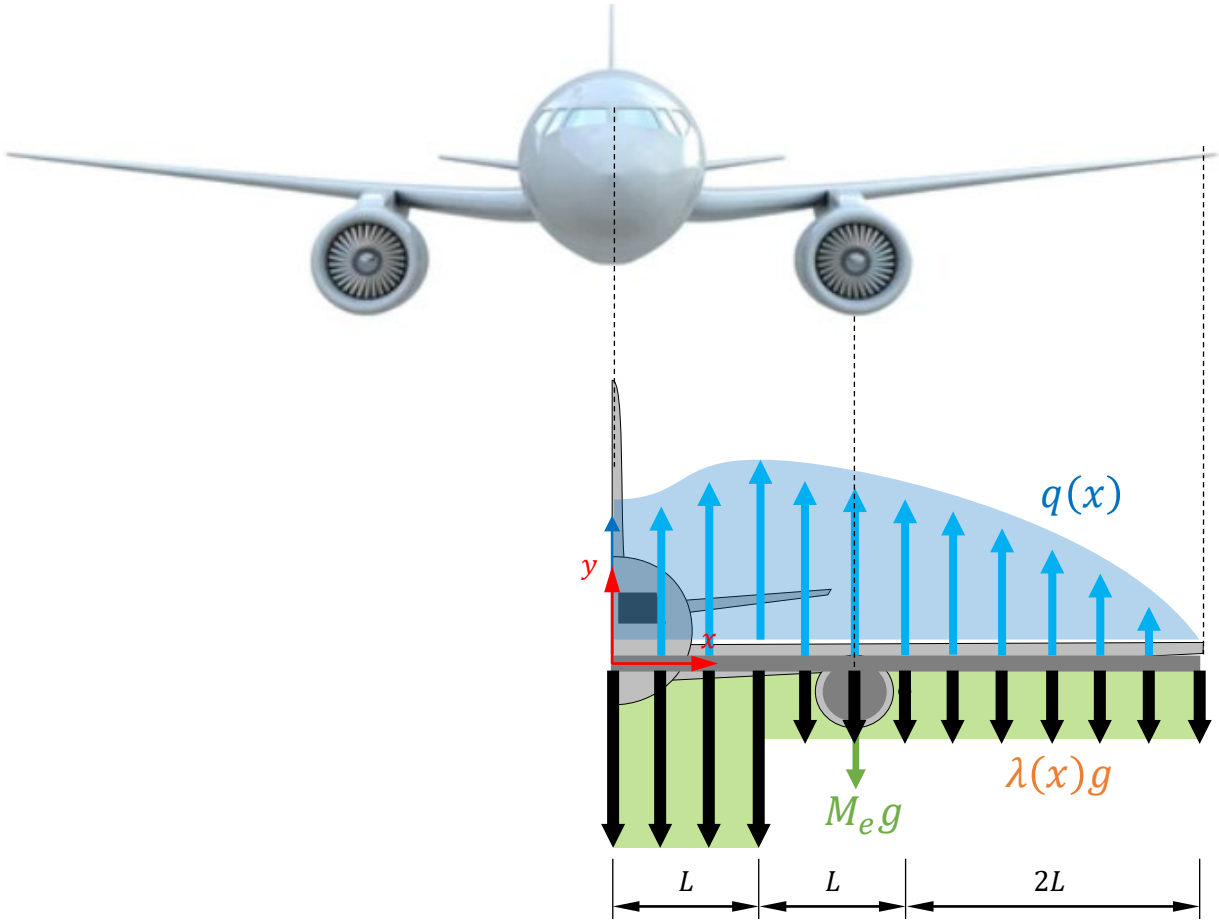
Structural analysis of an aircraft

- **Weight** generated by all the components of the aircraft
- **Lift**, the aerodynamic force provided by the air
- **Drag**, the air resistance or fluid resistance
- **Thrust** provided by the machine
- **Other forces**: Maneuvers, Gusts, Cabin pressure, Landing.



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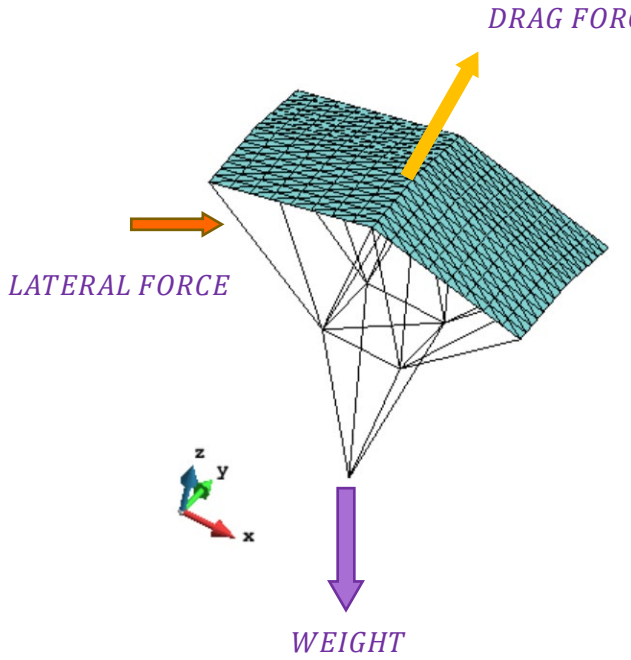
Structural analysis of a parachute

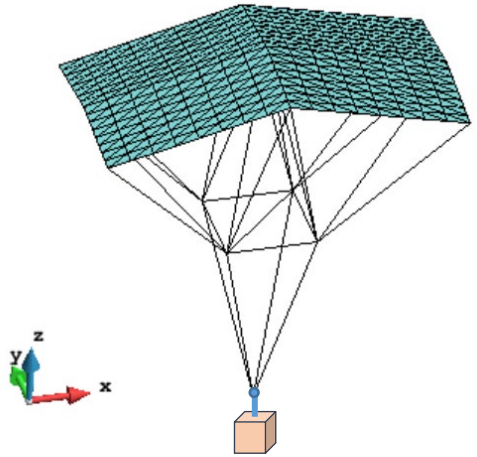
- **Weight** generated by all the components of the parachute

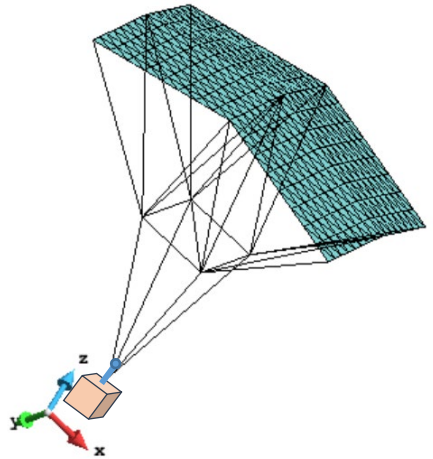
- **Drag**, the air resistance or fluid resistance

- **Inertial forces** generated by all the components of the parachute



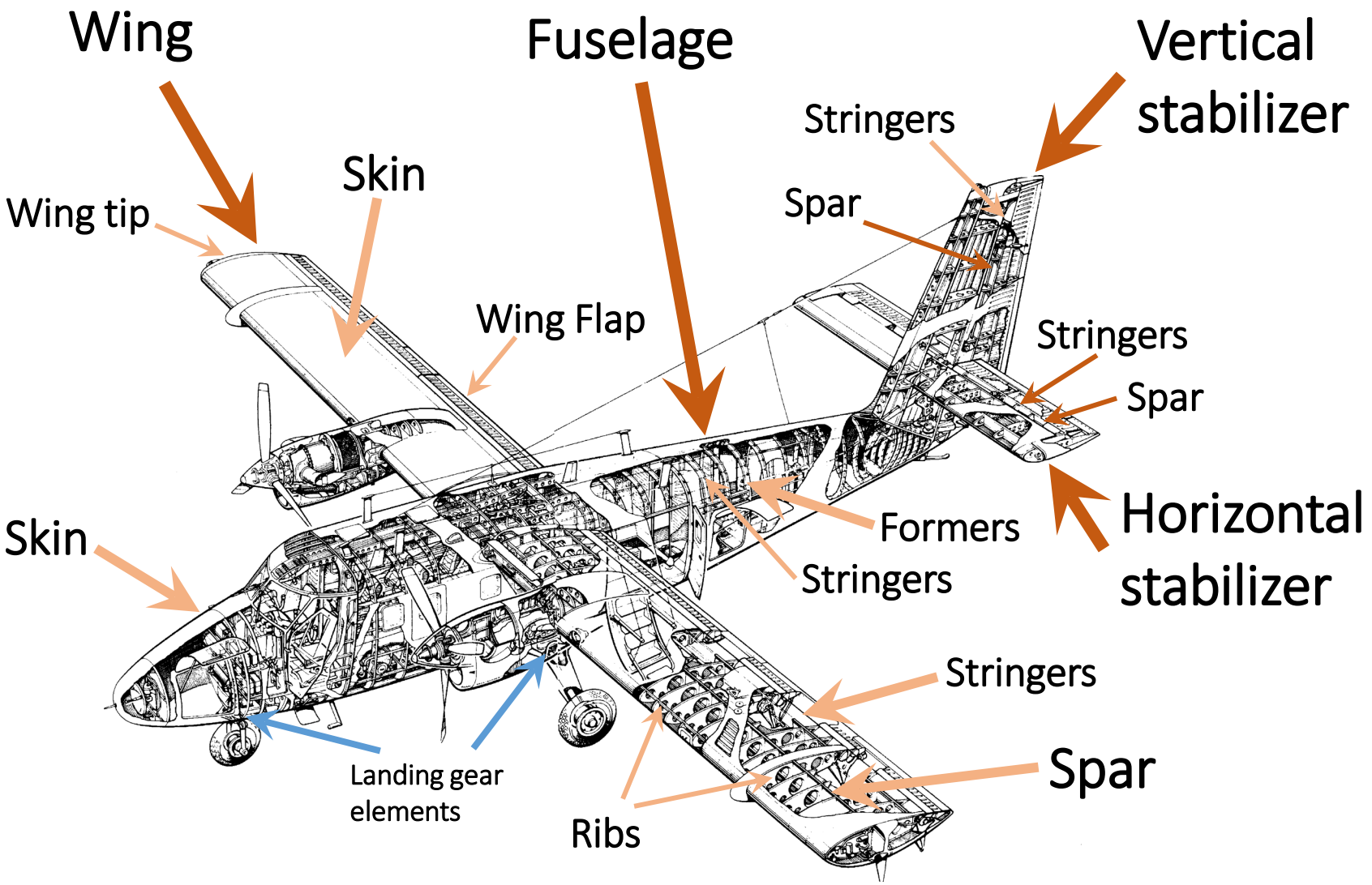






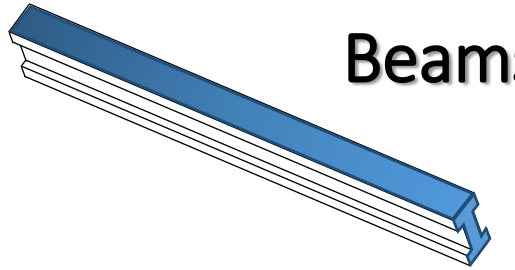
Structural elements in an airplane

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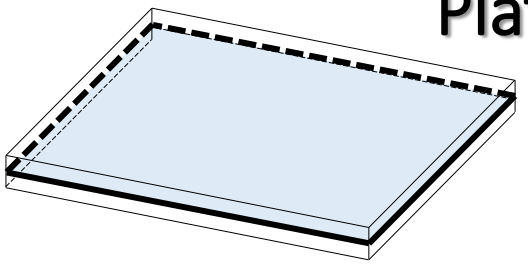


Proposed Structural idealization

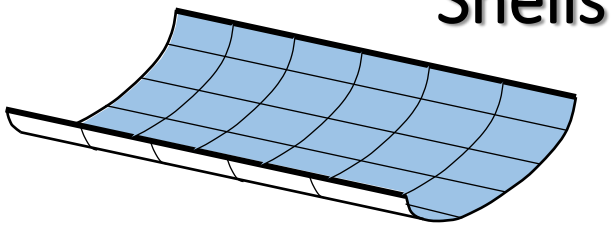
Beams



Plates

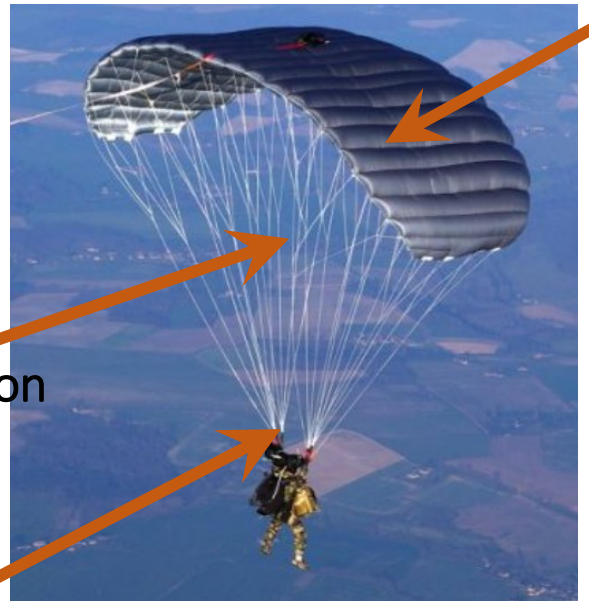


Shells



Struct. Elem. in a parachute and space structures

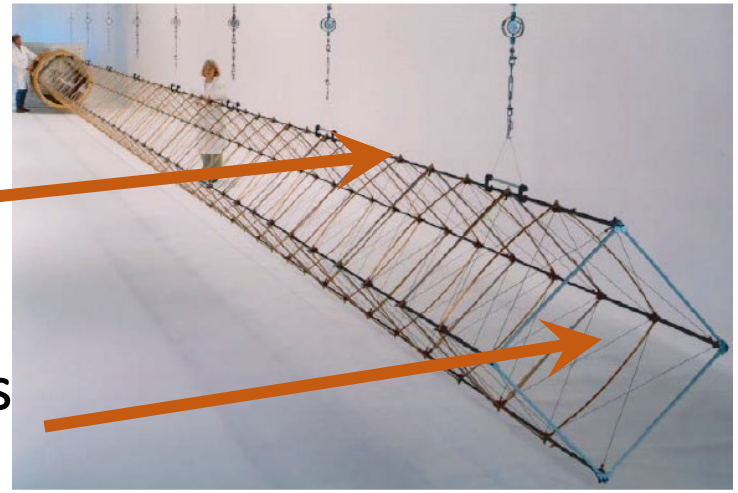
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Canopy

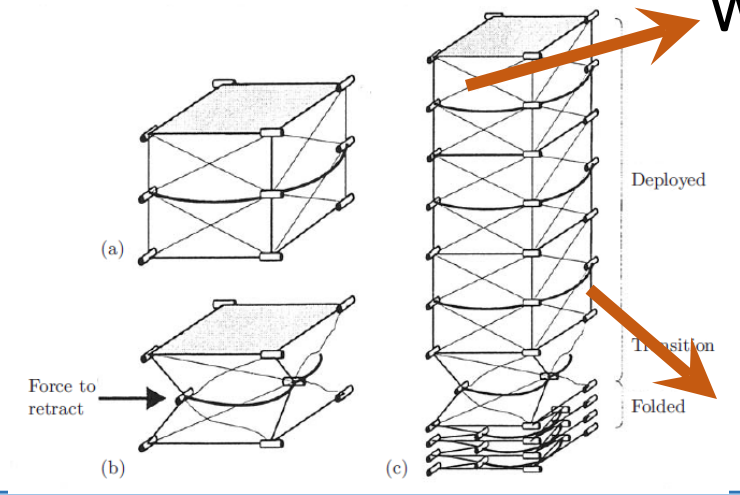
Suspension lines

links



Bars

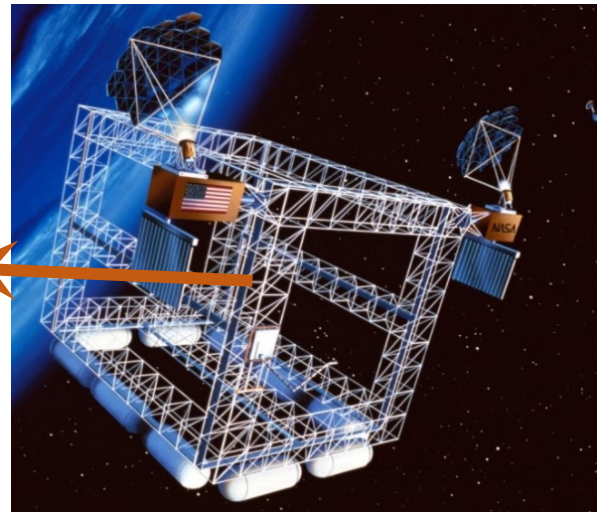
Diagonals or cables



Wires

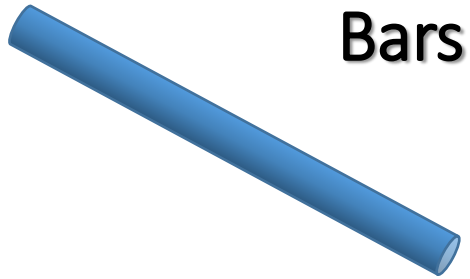
Truss struct.

Bars

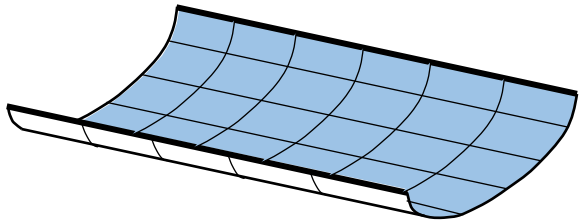


Structural idealization

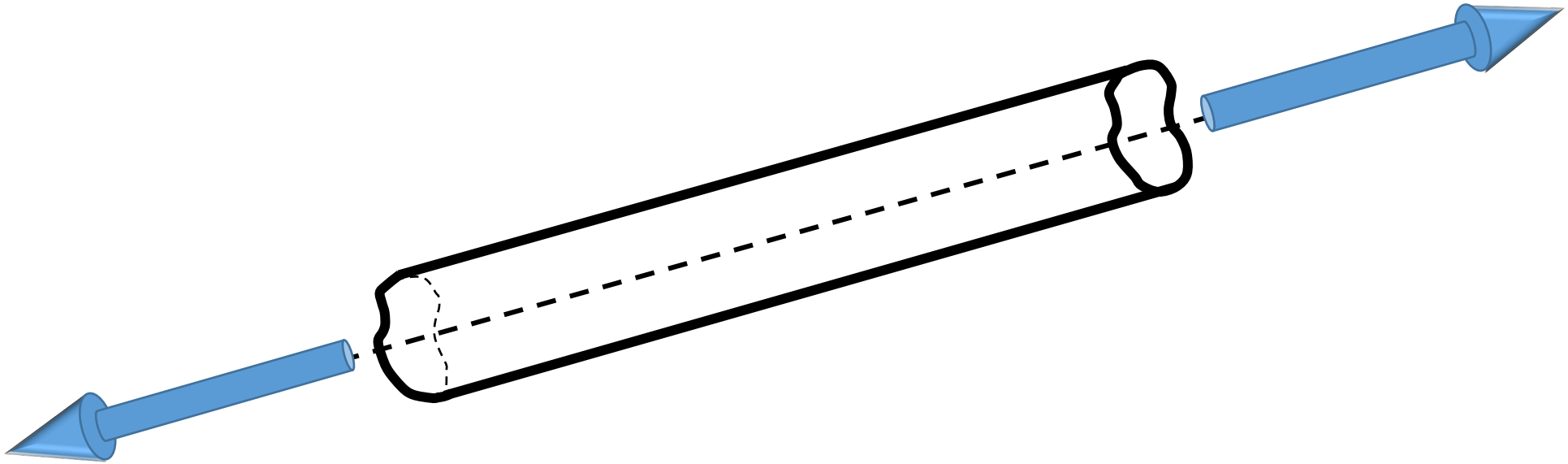
Bars

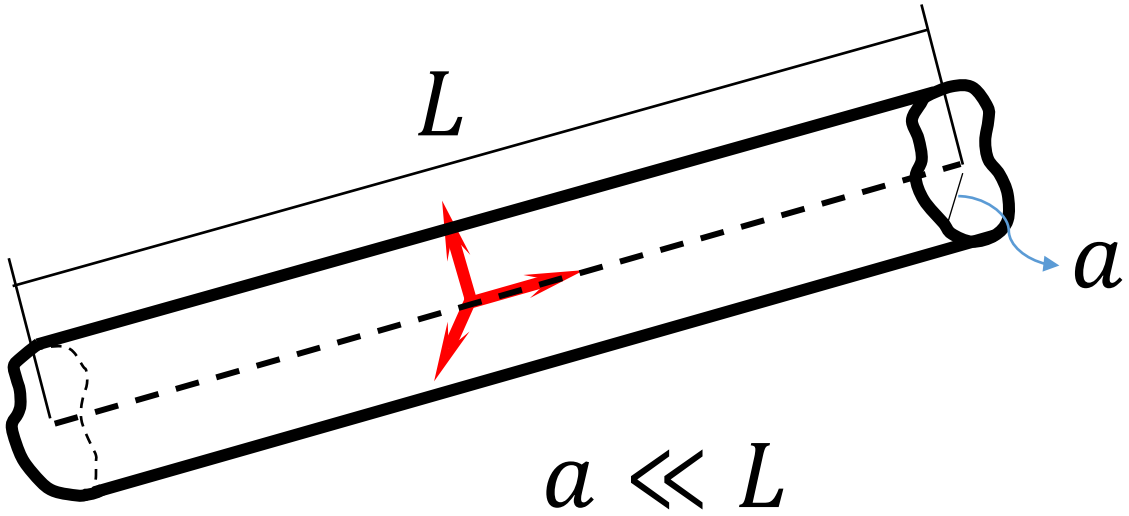


Shells

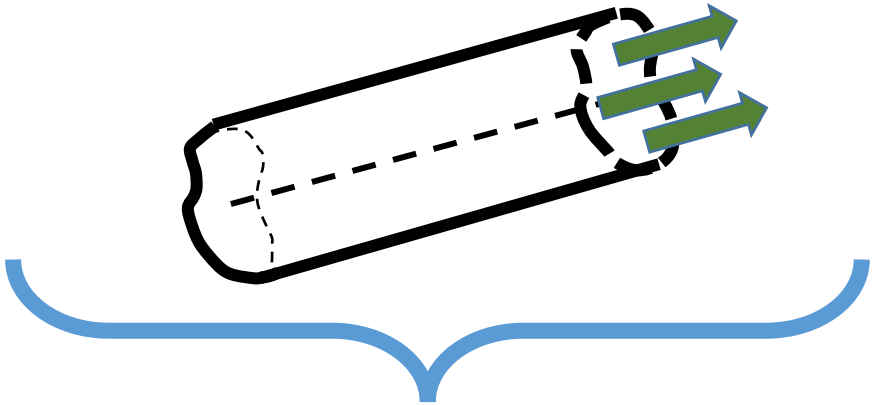


Structural idealization: BARS



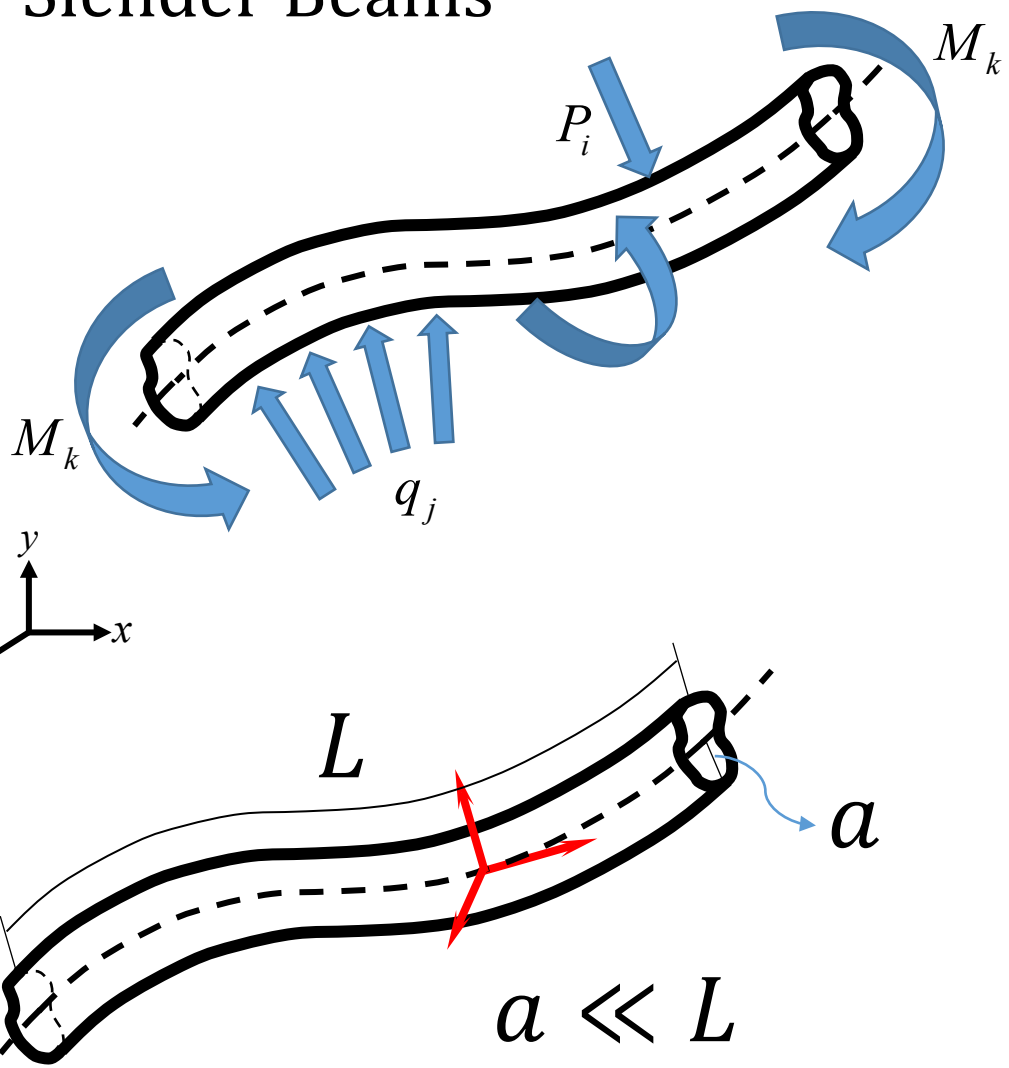


$a \ll L$

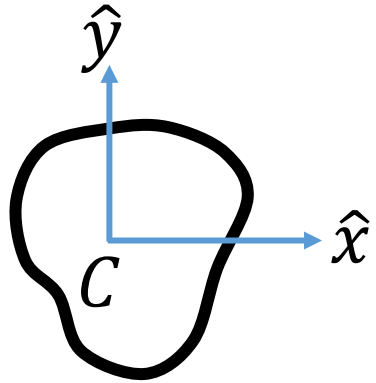


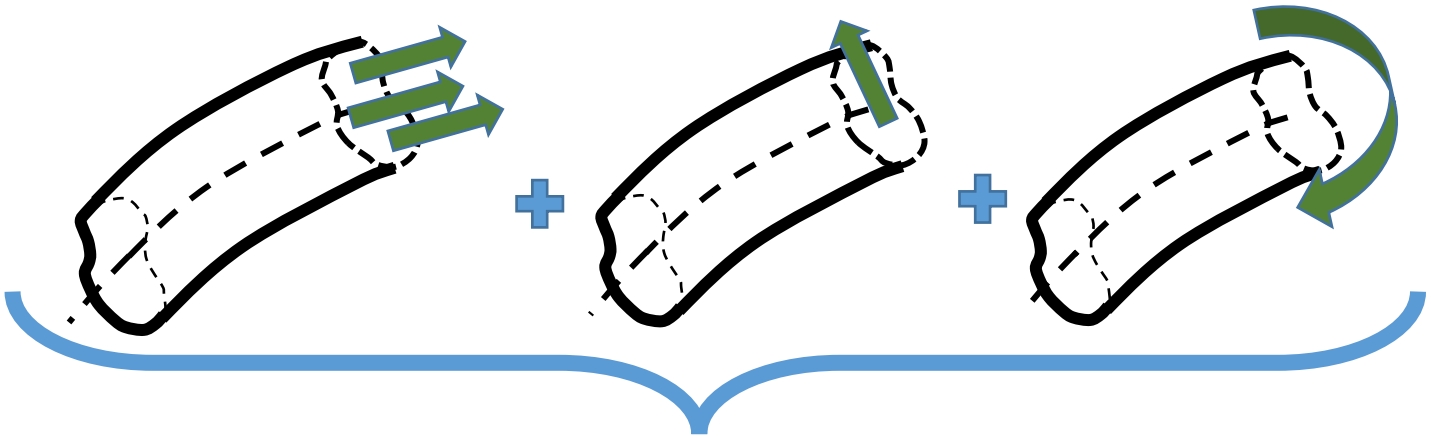
Structural idealization: BEAMS

Slender Beams

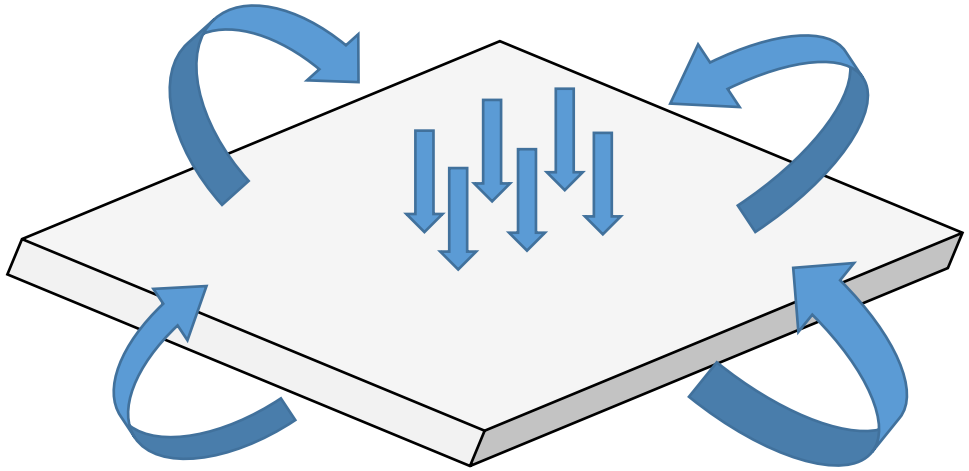


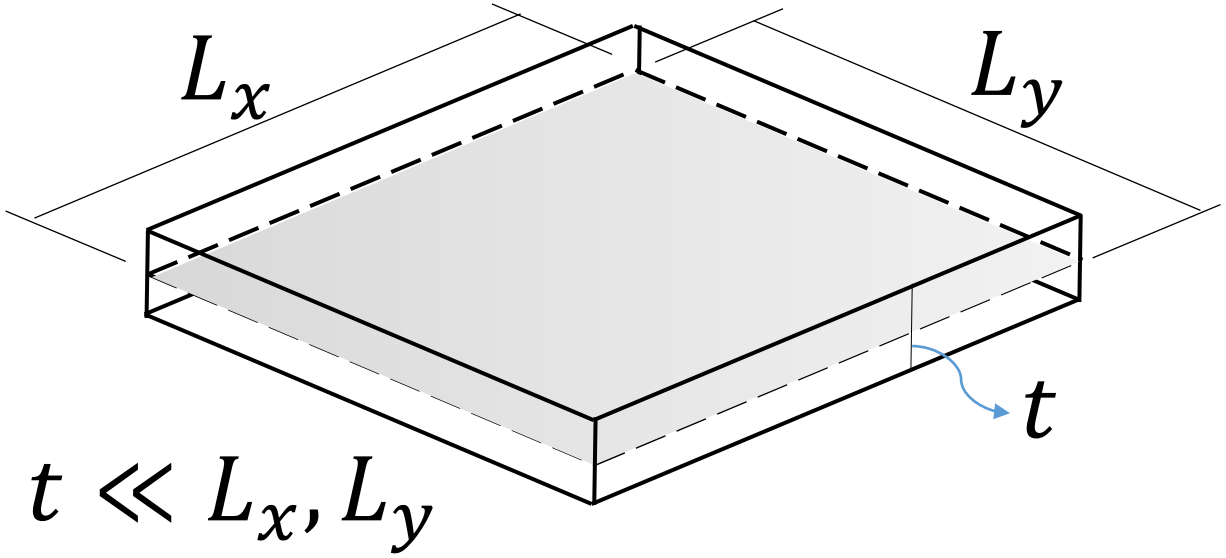
Beam cross section

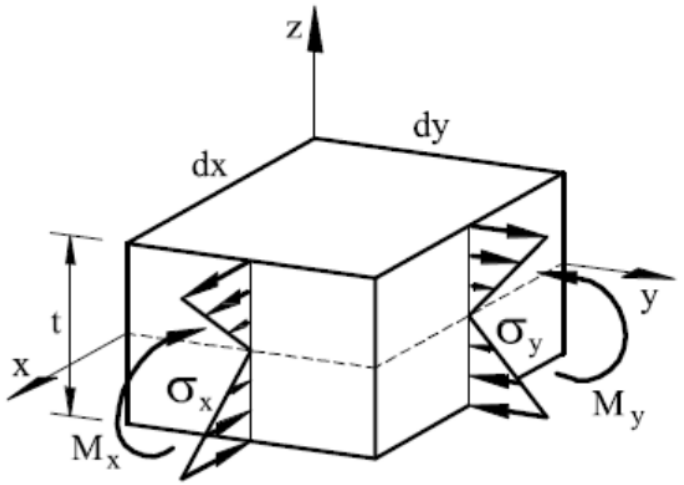


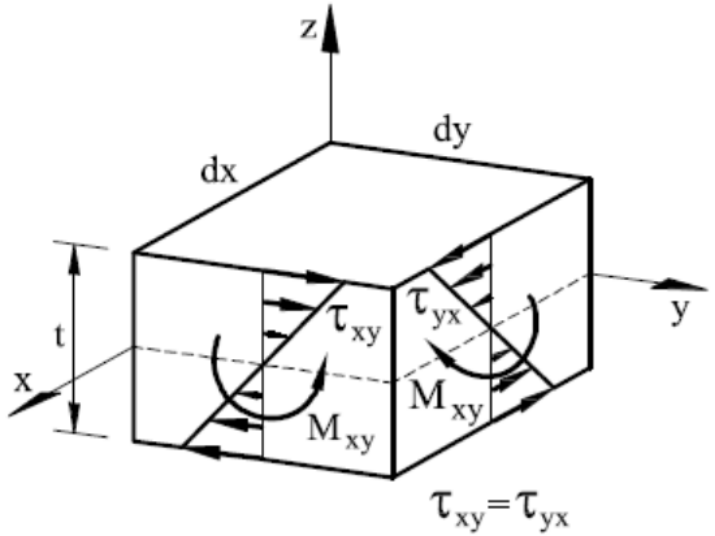


Structural idealization: PLATES



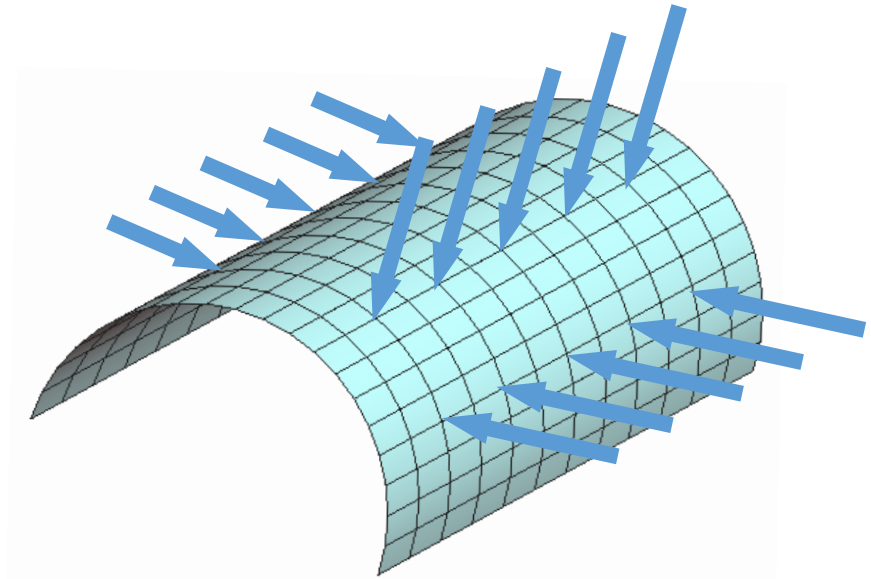


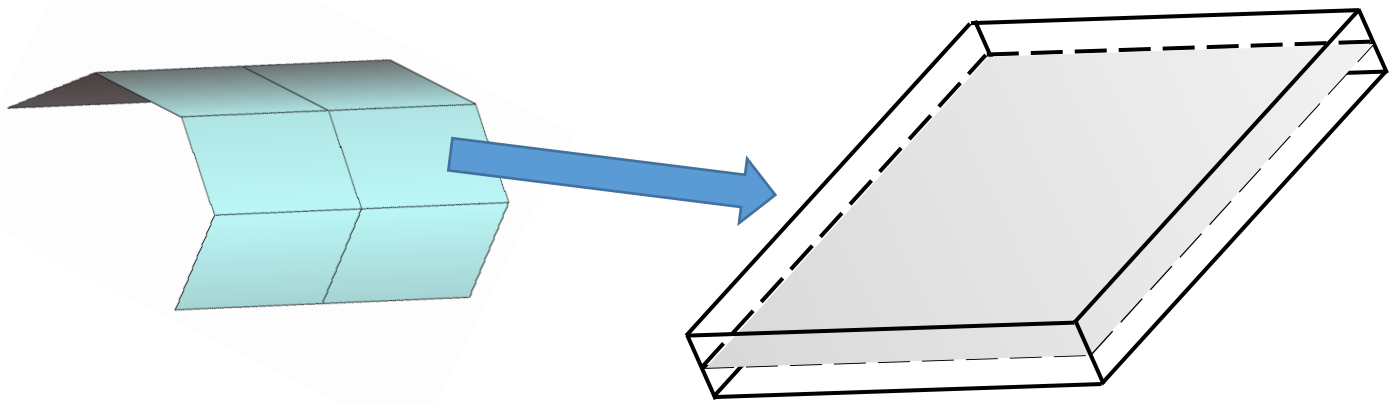


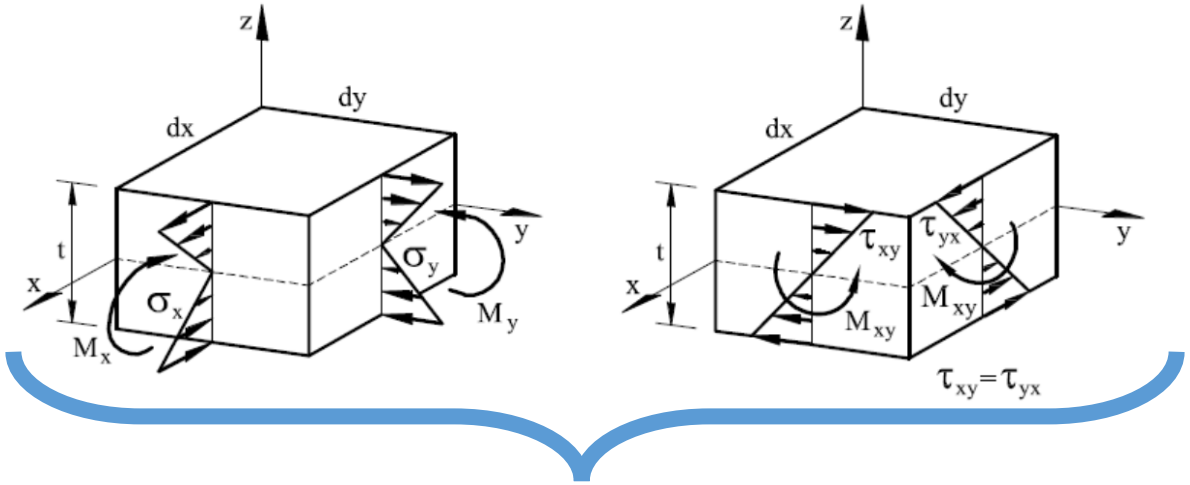


Structural idealization: SHELLS



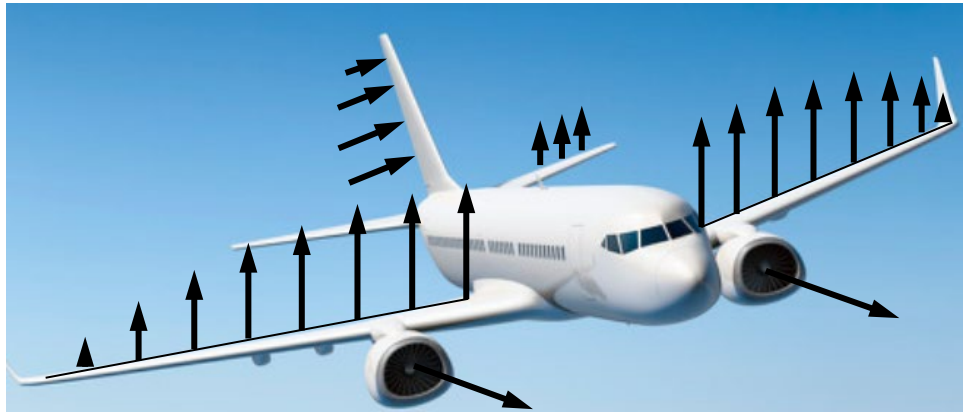






Structural problem to be solved

Given the external forces acting on the aircraft, the aim is to **find the internal loads and its distribution**



We need to satisfy several equations:

- ✓ Movement equations (Newton' law)
- ✓ Constitutive equations (Homogeneous of composite material)
- ✓ Boundary and initial conditions

Solution Methods:

- ✓ Analytical methods: equilibrium, hyper-static meth, Moments,…)
- ✓ **Numerical methods: FEM + composite analysis**

Objective

Development of numerical tools, based on the FEM, for the *structural analysis* and *optimization* of different aeronautical parts.

Different structural topologies and new concepts will be considered:

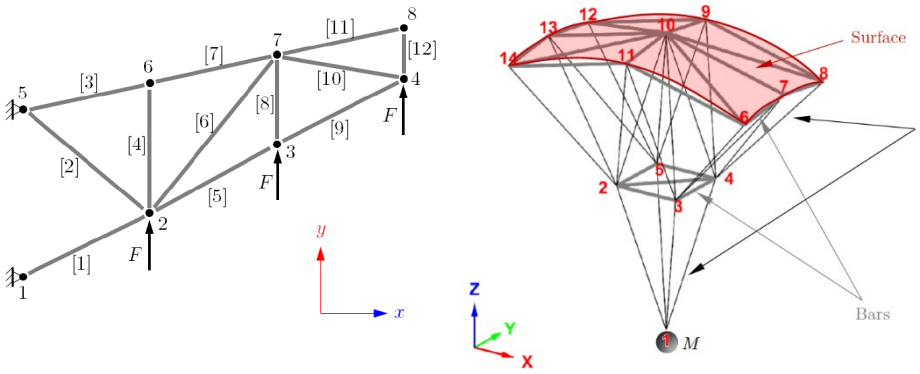
- Bar element (review)
- Beam element (Timoshenko theory)
- Plate element (Reissner-Mindlin theory)
- Shell element
- Non-linear beam element (**large strains and large displacements**)
- Introduction to topology optimization

Materials

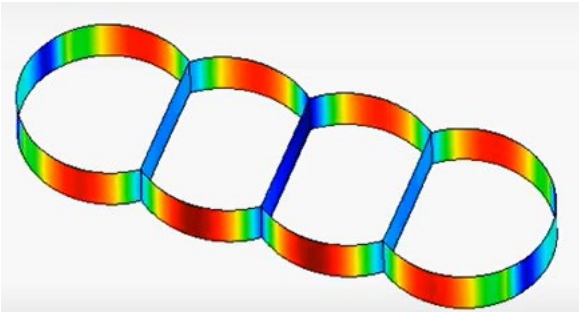
- Homogeneous beams, plates, and shells
- Composites laminated beams, plates, and shells

Table of contents

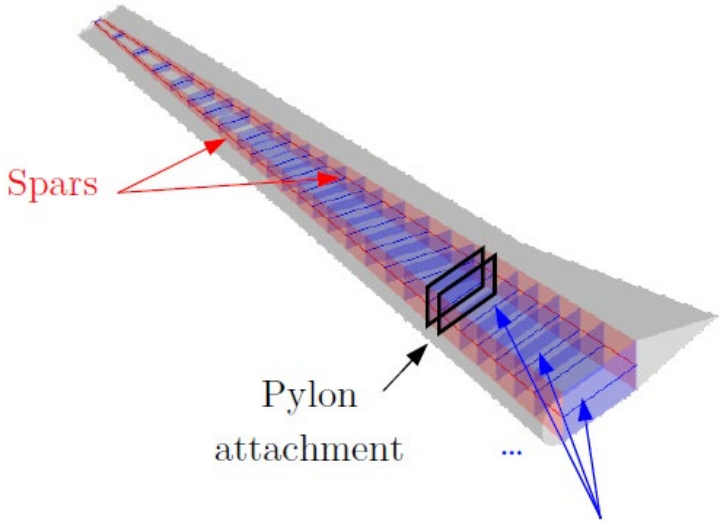
Bar element (review)

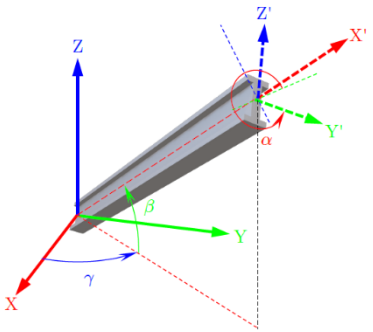


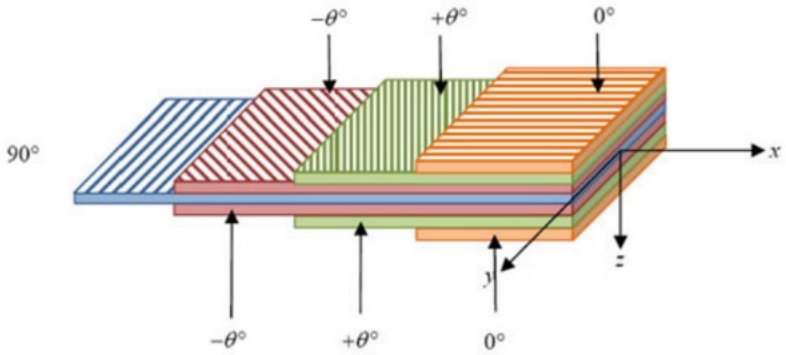
The base of the FEM



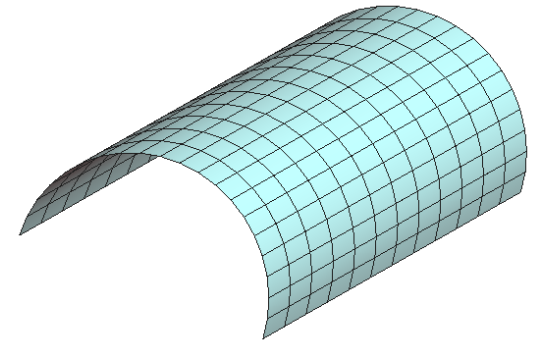
Timoshenko Beams. Composites

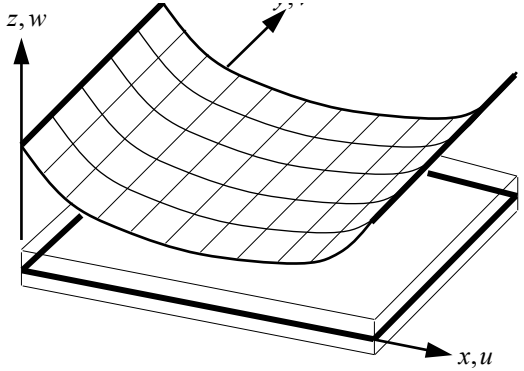


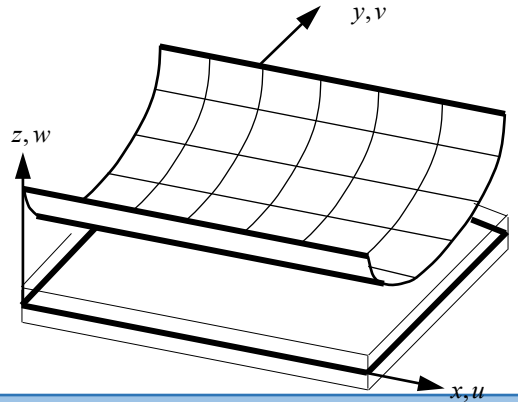




Non linear beams







Large strains and large displacements (Geometrically nonlinear analysis)



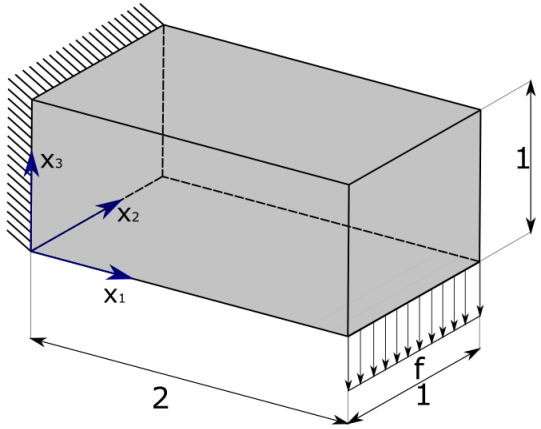
Glider (sailplane)



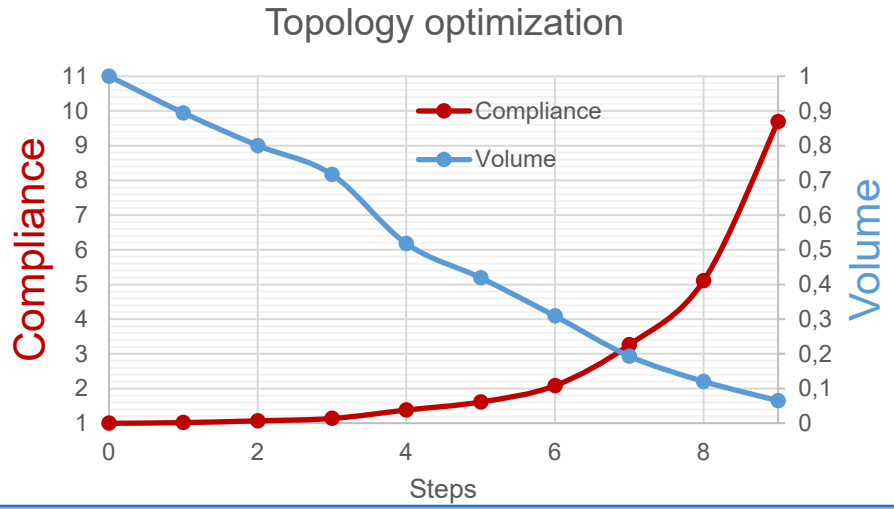


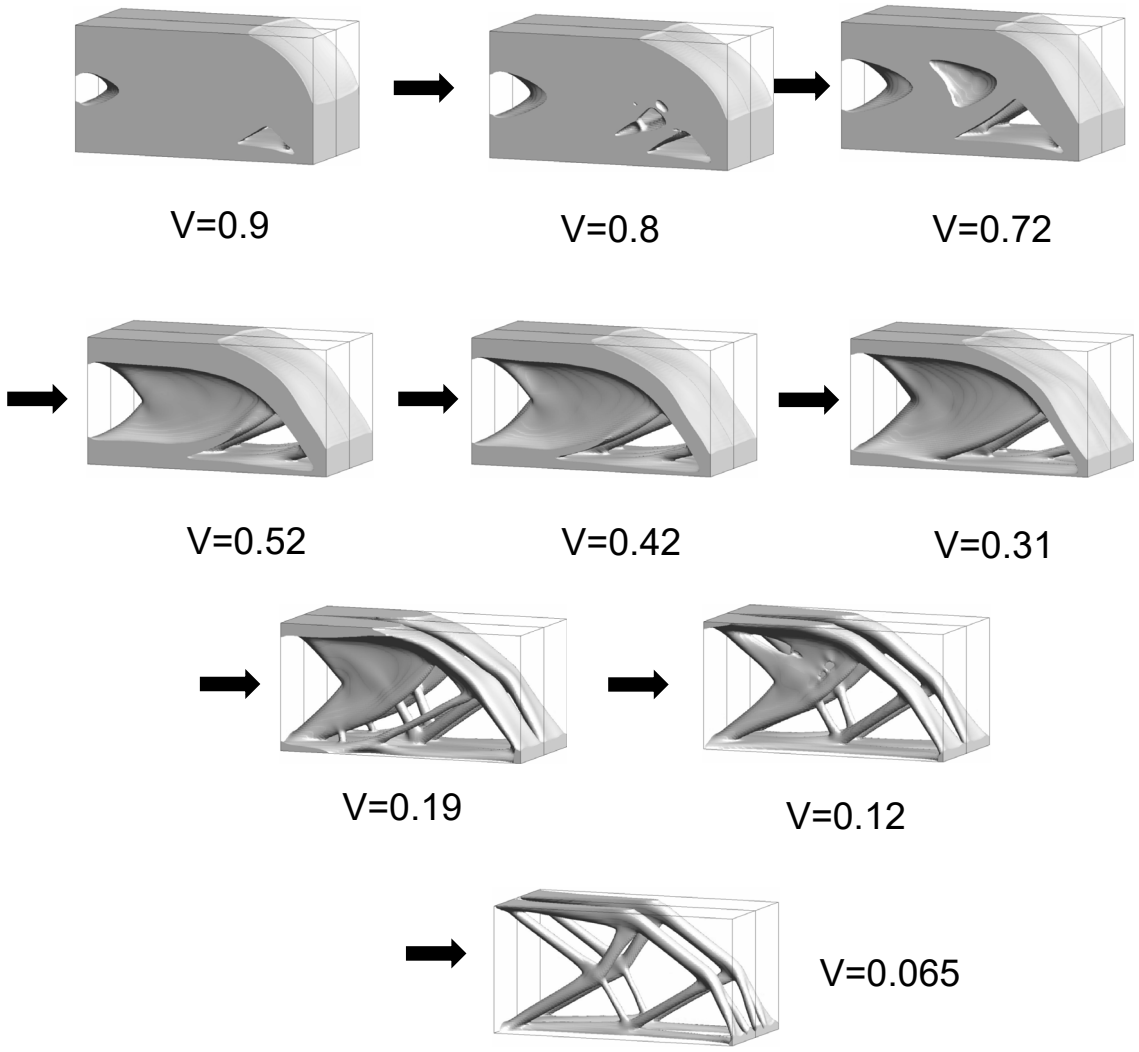
Comments about optimization techniques

Minimum mean compliance: Cantilever 3D



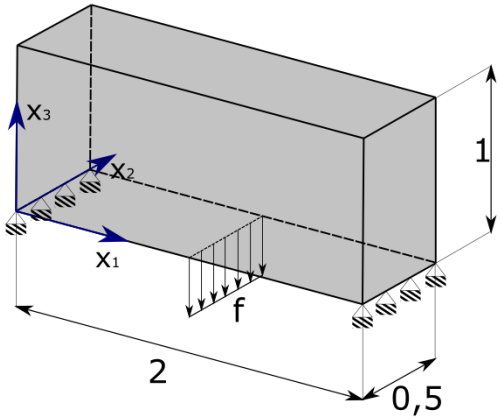
Initial (design) domain



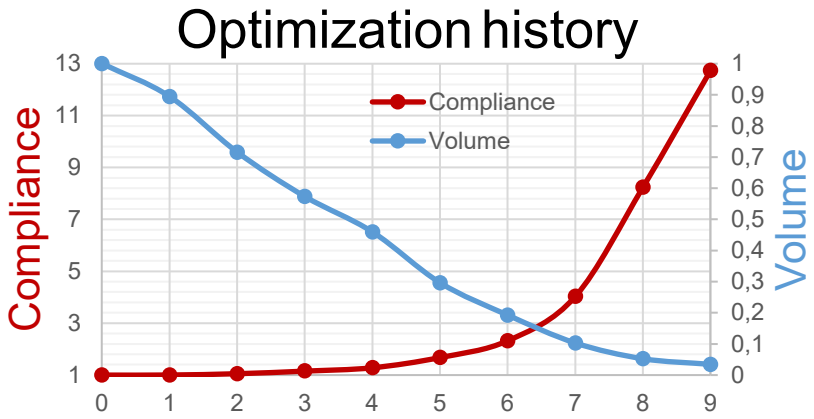


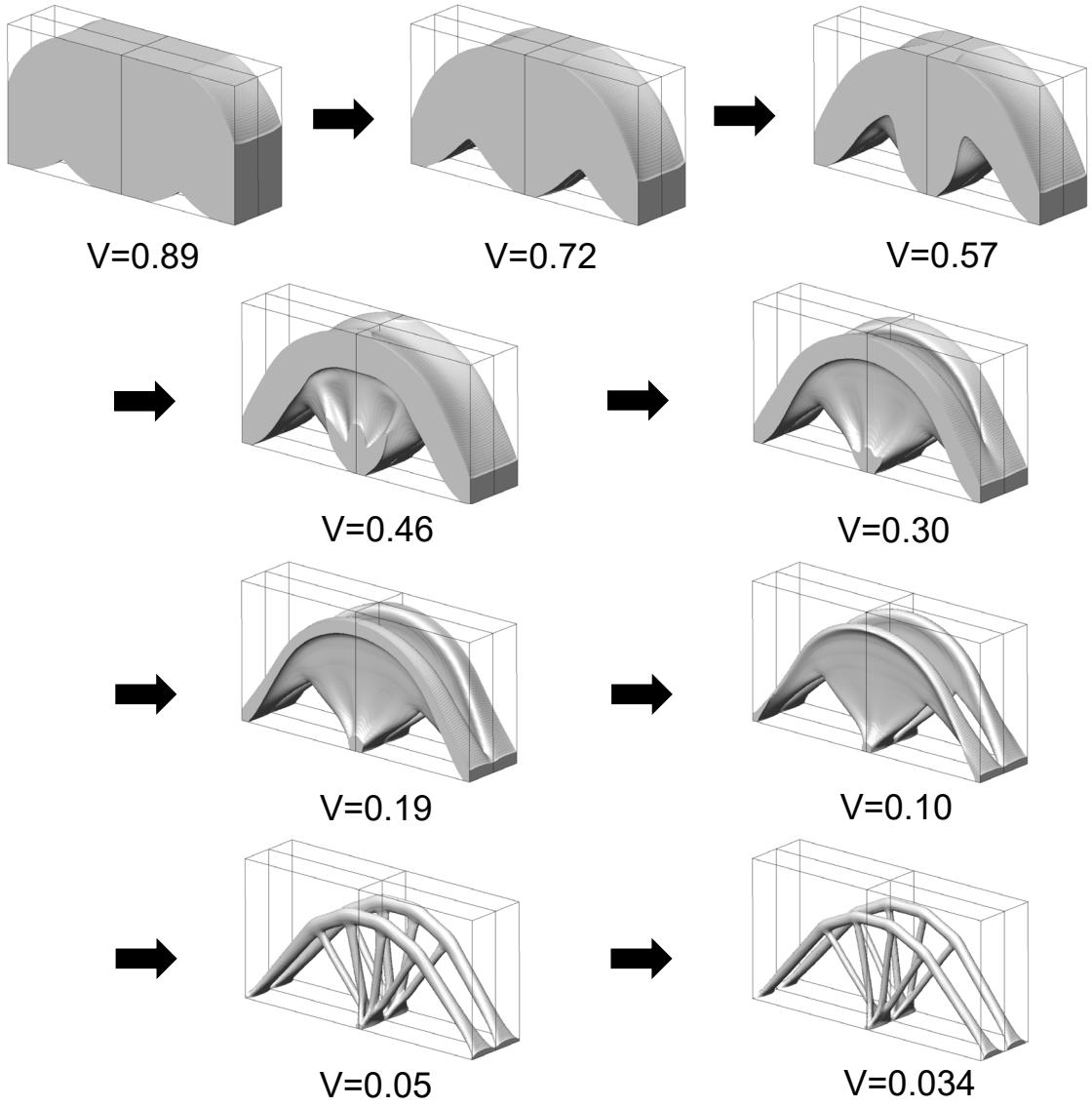
Minimum mean compliance: Michell structure 3D

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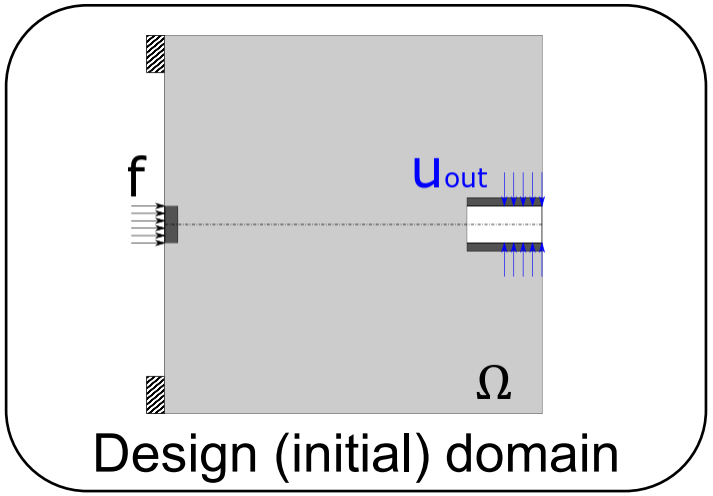
Fixed (design) domain

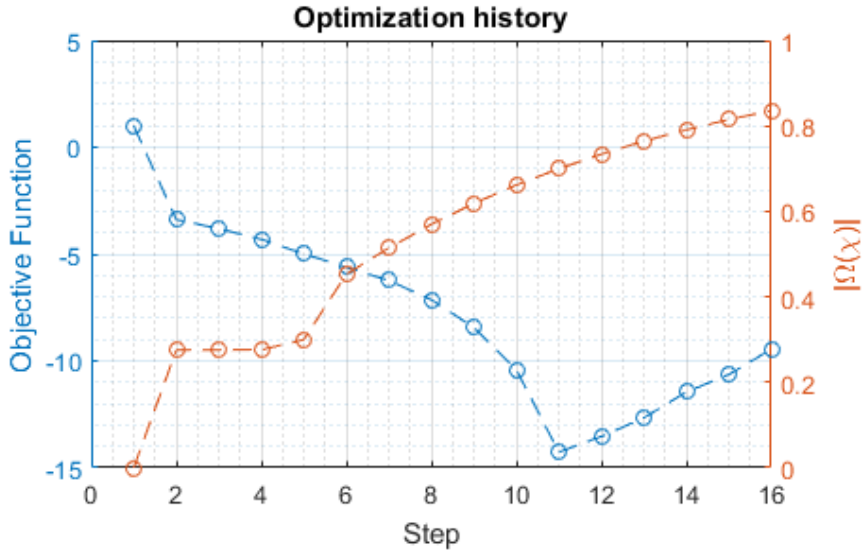


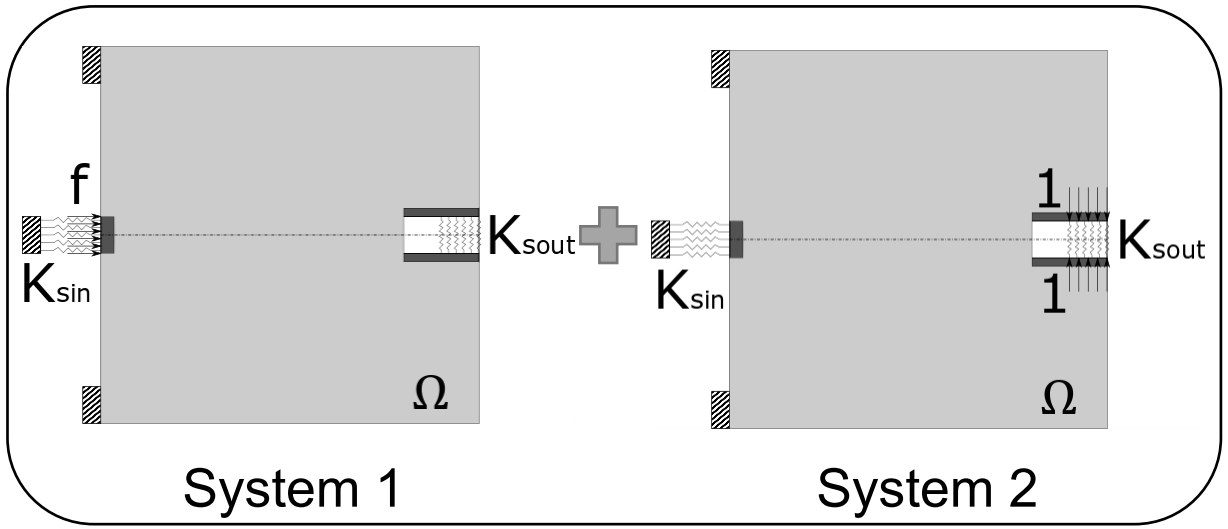


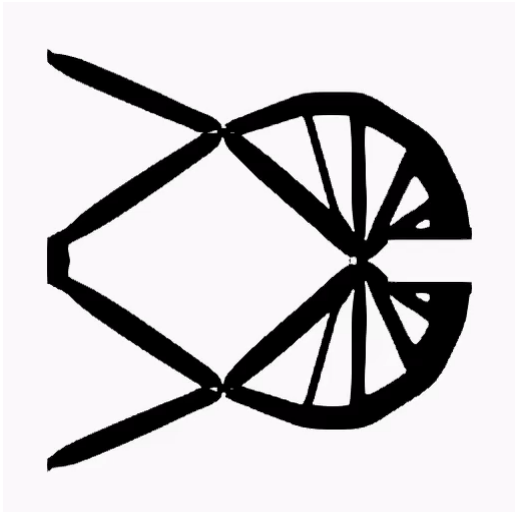
Compliant mechanism: Gripper 2D

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Qualification system of the second part of the course

- **Five assignments will be proposed along this second part of course**

1. 1D Finite Element Method: linear and quadratic element.
2. 1D Beam problem: mode analysis (Timoshenko beam theory)
3. 2D Plate problem: uniform distributed load test (Reissner-Mindlin)
4. 1D truss structure in large strains.
5. 2D optimization problems.

- **A final project (closely related to assignments 2 and 3).**

30% Assignments (3)

- Only 3 assignments are mandatory
- The first assignment is compulsory
- The other two are chosen from among the others

20% Final project.

- The defense of the work (project and assignments) will take place during the period of the final exams in an oral evaluation version (January 8 – January 19).
- For the defense, remember to have the MATLAB codes and the reports ready to show.
- There is no need to prepare a PowerPoint presentation of the work.
- The assignments can be done in groups of maximum 2 people

With the three assignments and the final project, a maximum grade of 9.0 is awarded. To opt to an "Honorable mention (10)", on this part, all the assignments should be completed

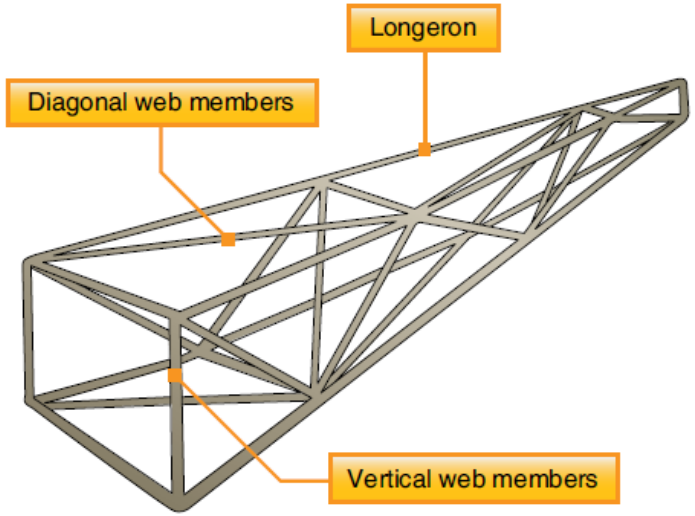
Bar element

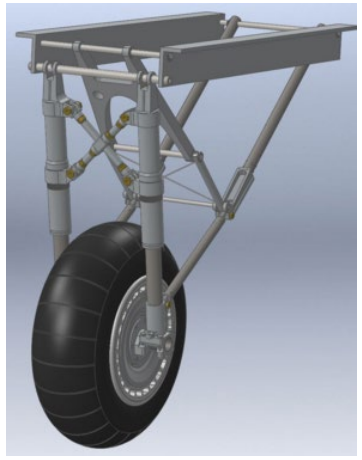
(review)

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Bar elements

A collection of bar elements defines a truss structure

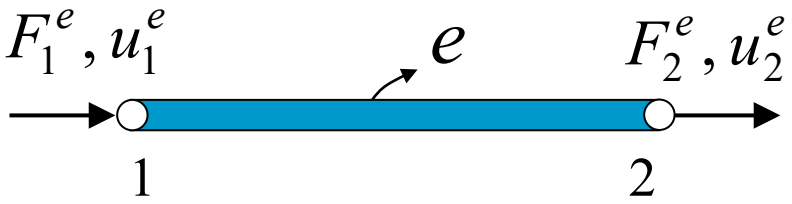




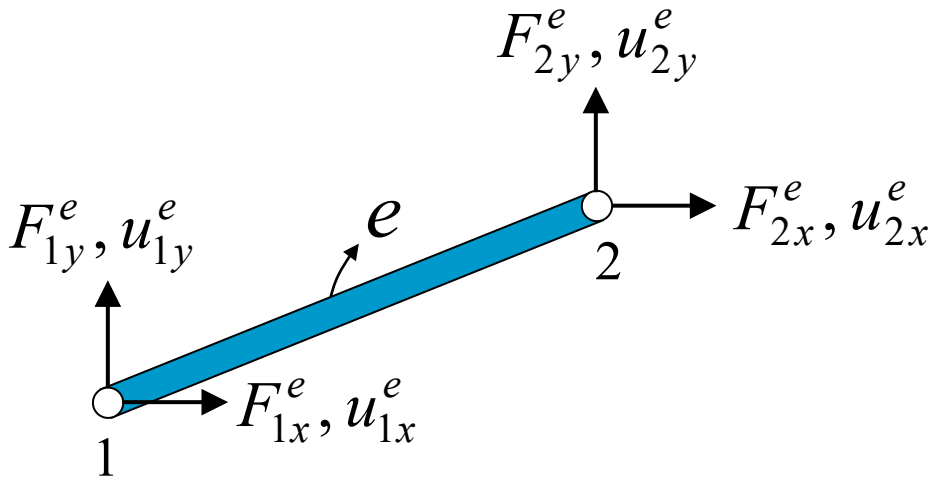
- A **bar element** is assumed to be **very thin** (slender element) They have **negligible** resistance to **torsion**, **bending** and **shear**
- The only internal forces are **axial (internal) forces**

Element description

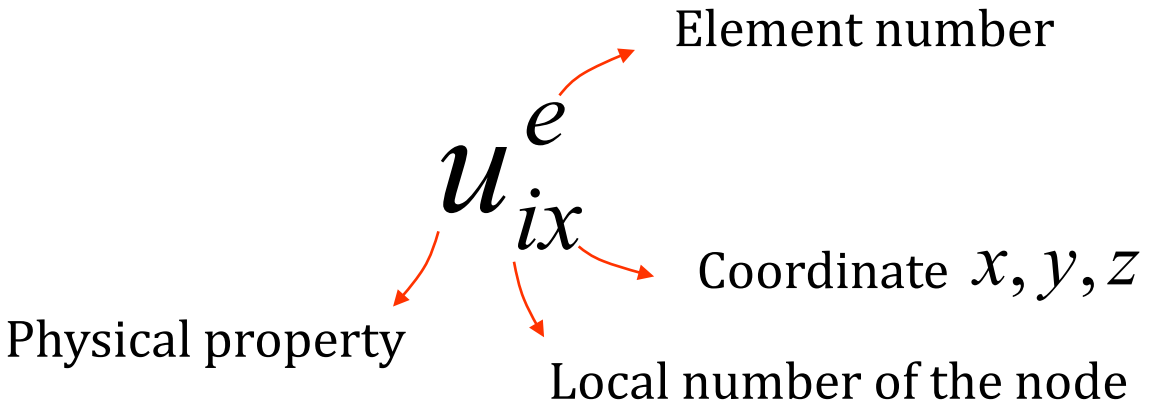
1-D



2-D

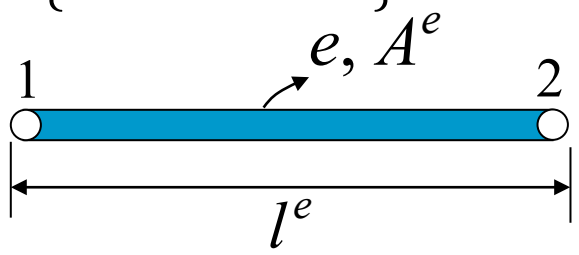


Comments about notation:



Behavior of a single 1-D bar element

1) Geometry $\left\{ A^e, l^e, x_1^e, x_2^e \right\}$



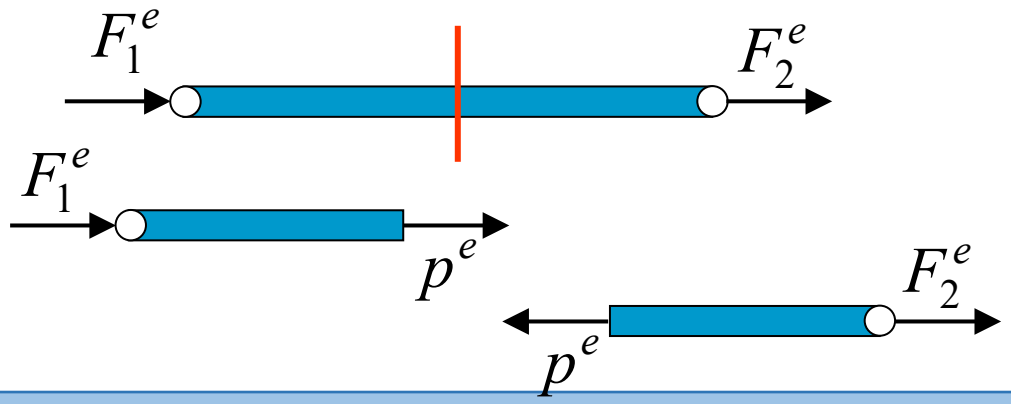
Different cross sections





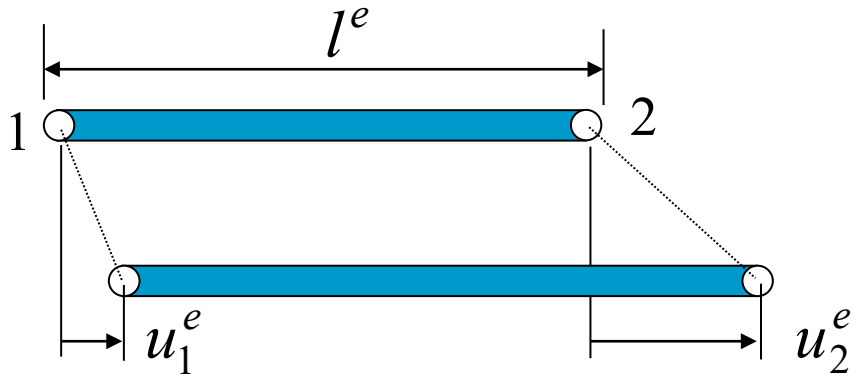
2) Material $\left\{ E^e, \nu^e \right\}$

3) Internal force $\left\{ p^e \right\}$



Axial stress $\sigma^e = \dfrac{p^e}{A^e}$

4) Displacement and Strain



$$l^e = \left| x_2 - x_1 \right| \qquad x_2 > x_1$$

$$u^e(x) = \frac{u_2^e - u_1^e}{l^e}(x - x_1) + u_1^e$$

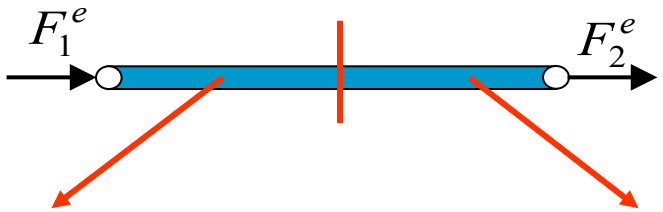
$$\varepsilon^e = \frac{\partial u}{\partial x} = \frac{u_2^e - u_1^e}{l^e}$$

5) Hooke's law

$$\sigma^e = E^e \varepsilon^e = E^e \frac{(u_2^e - u_1^e)}{l^e}$$

$$\Rightarrow \quad p^e = A^e \sigma^e = A^e E^e \frac{(u_2^e - u_1^e)}{l^e}$$

6) Equilibrium $F_1^e + F_2^e = 0$



$F_1^e + p^e(x) = 0$ $\qquad -p^e(x) + F_2^e = 0$

7) Forces and element displacements

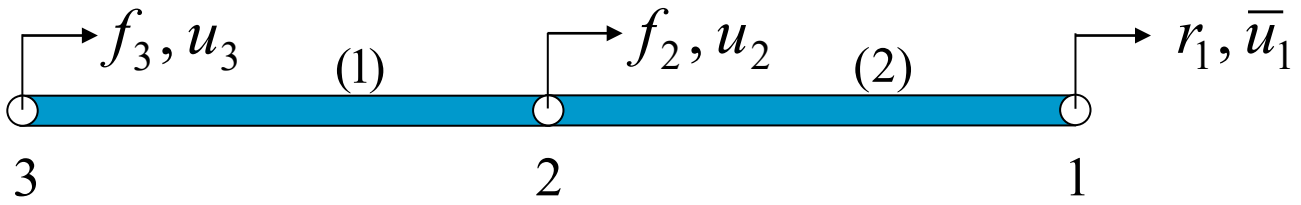
$$p^e = F_2^e \rightarrow F_2^e = \frac{A^e E^e}{l^e}(u_2^e - u_1^e) =: k^e (u_2^e - u_1^e)$$

$$F_1^e = -F_2^e = \frac{A^e E^e}{l^e}(u_1^e - u_2^e) =: k^e (u_1^e - u_2^e)$$

$$\underbrace{\begin{bmatrix} F_1^e \\ F_2^e \end{bmatrix}}_{\mathbf{F}^e} = \underbrace{\begin{bmatrix} k^e & -k^e \\ -k^e & k^e \end{bmatrix}}_{\mathbf{K}^e} \underbrace{\begin{bmatrix} u_1^e \\ u_2^e \end{bmatrix}}_{\mathbf{d}^e}$$

$$\Rightarrow \mathbf{F}^e = \mathbf{K}^e \mathbf{d}^e$$

General problem: two elements



$$\mathbf{F}^{(1)} = \mathbf{K}^{(1)}\mathbf{d}^{(1)}$$

$$\mathbf{F}^{(2)} = \mathbf{K}^{(2)}\mathbf{d}^{(2)}$$

$$\begin{bmatrix} F_1^e \\ F_2^e \end{bmatrix} = \begin{bmatrix} k^e & -k^e \\ -k^e & k^e \end{bmatrix} \begin{bmatrix} u_1^e \\ u_2^e \end{bmatrix} \qquad e = 1,2$$

$$\underbrace{\begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \\ -k^{(2)} & k^{(2)} + k^{(1)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}}_{\mathbf{d}} = \underbrace{\begin{bmatrix} r_1 \\ f_2 \\ f_3 \end{bmatrix}}_{\mathbf{f}}$$

Solution of the global system of equations

- Remember that: $\{f_2, f_3\}$ and $\{\bar{u}_1\}$ are known !

- The system equation is then partitioned as follows:

$$\left[\begin{array}{c|cc} k^{(2)} & -k^{(2)} & 0 \\ \hline -k^{(2)} & k^{(2)}+k^{(1)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{array}\right]\left[\begin{array}{c} \bar{u}_1 \\ \hline u_2 \\ u_3 \end{array}\right]=\left[\begin{array}{c} r_1 \\ \hline f_2 \\ f_3 \end{array}\right]$$

- The system can be written in the following block matrix form:

$$\begin{bmatrix} \mathbf{K}_{RR} & \mathbf{K}_{RL} \\ \mathbf{K}_{LR} & \mathbf{K}_{LL} \end{bmatrix}\begin{bmatrix} \mathbf{d}_R \\ \mathbf{d}_L \end{bmatrix}=\begin{bmatrix} \mathbf{f}_R \\ \mathbf{f}_L \end{bmatrix}$$

$$\mathbf{d}_R = \left[\bar{u}_1\right] \qquad \mathbf{f}_R = \left[r_1\right] \qquad \mathbf{K}_{RR} = \left[k^{(2)}\right] \qquad \mathbf{K}_{RL} = \left[-k^{(2)} \quad 0\right]$$

$$d_L = \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} \qquad \mathbf{f}_L = \begin{bmatrix} f_2 \\ f_3 \end{bmatrix} \qquad \mathbf{K}_{LR} = \begin{bmatrix} -k^{(2)} \\ 0 \end{bmatrix} \qquad \mathbf{K}_{LL} = \begin{bmatrix} k^{(2)} + k^{(1)} & -k^{(1)} \\ -k^{(1)} & k^{(1)} \end{bmatrix}$$

- After multiplying by blocks, we obtain :

$$\mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L = \mathbf{f}_R$$

$$\mathbf{K}_{LR}\mathbf{d}_R + \mathbf{K}_{LL}\mathbf{d}_L = \mathbf{f}_L$$

-finally, displacements and reactions are obtained:

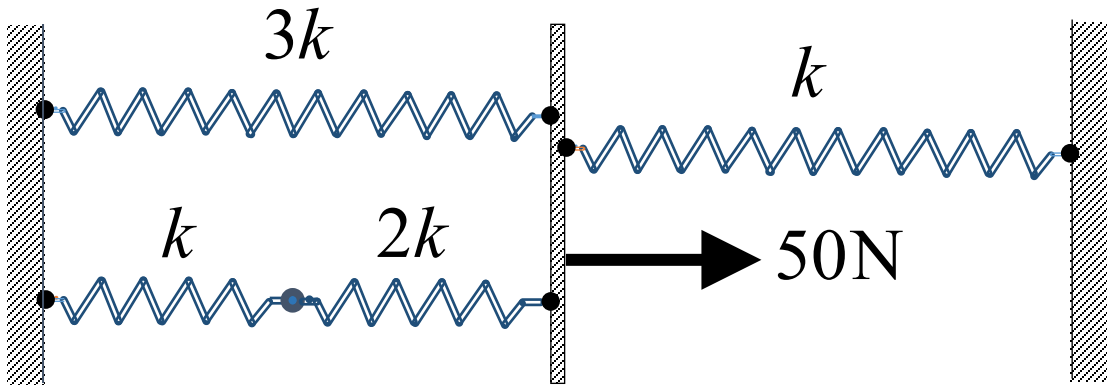
$$\mathbf{d}_L = \mathbf{K}_{LL}^{-1}\left(\mathbf{f}_L - \mathbf{K}_{LR}\mathbf{d}_R\right)$$

$$\mathbf{f}_R = \mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L$$

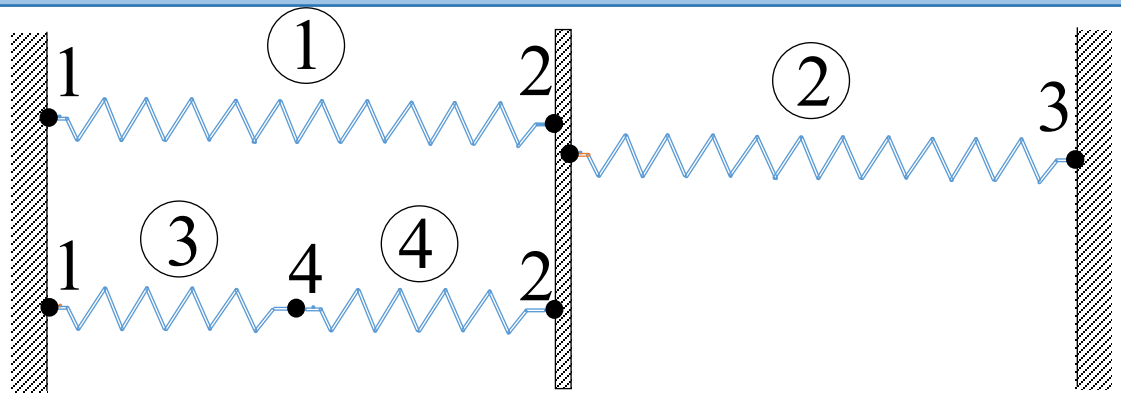
Example

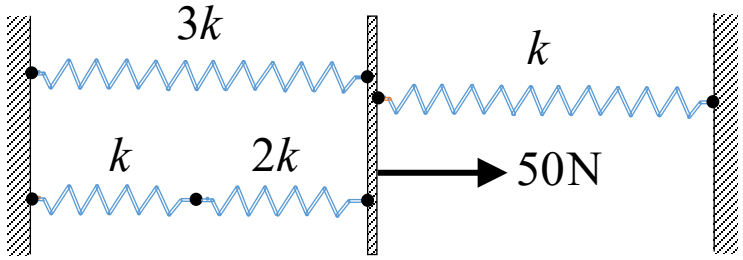
For the 1D structure, represented by the spring system given below,

a. Number the elements and nodes.
b. Assemble the global stiffness and force matrix.
c. Partition the system and solve the nodal displacements
d. Compute the reaction forces.



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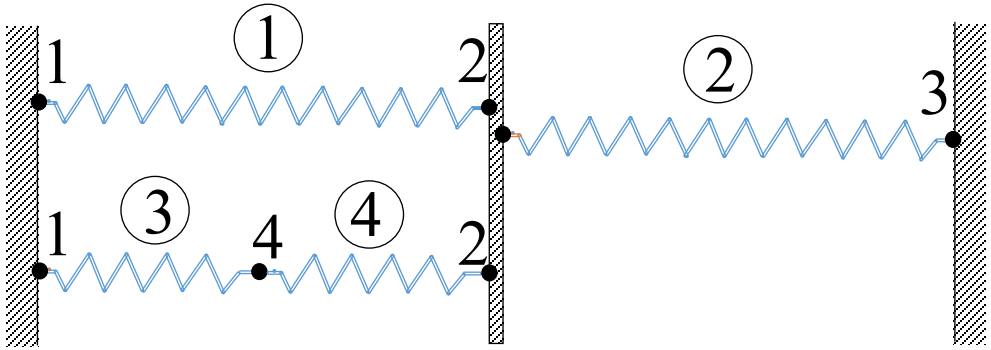


b1. Element stiffness matrices

$$\mathbf{K}^{(1)} = \begin{bmatrix} 3k & -3k \\ -3k & 3k \end{bmatrix} \qquad \mathbf{K}^{(2)} = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$$

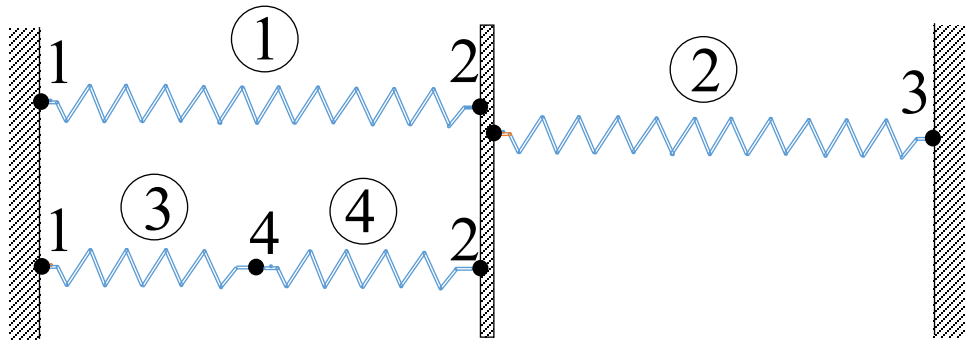
$$\mathbf{K}^{(3)} = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \qquad \mathbf{K}^{(4)} = \begin{bmatrix} 2k & -2k \\ -2k & 2k \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \\ 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$



$$\mathbf{K} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \end{bmatrix} \\ \begin{bmatrix} 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

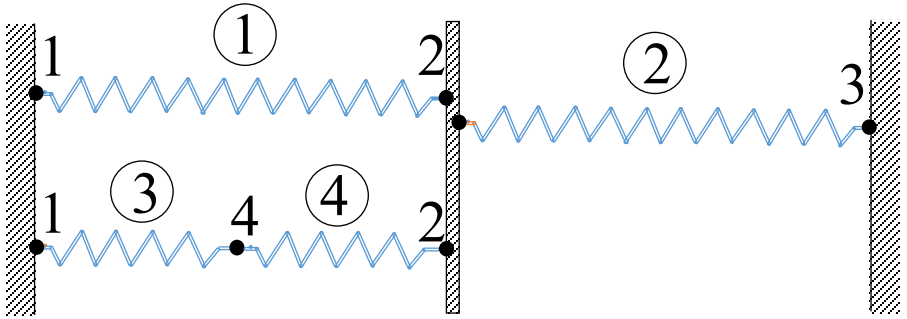


$$\mathbf{K} = \begin{bmatrix} 3k & -3k & 0 & 0 \\ -3k & 3k & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \\ 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

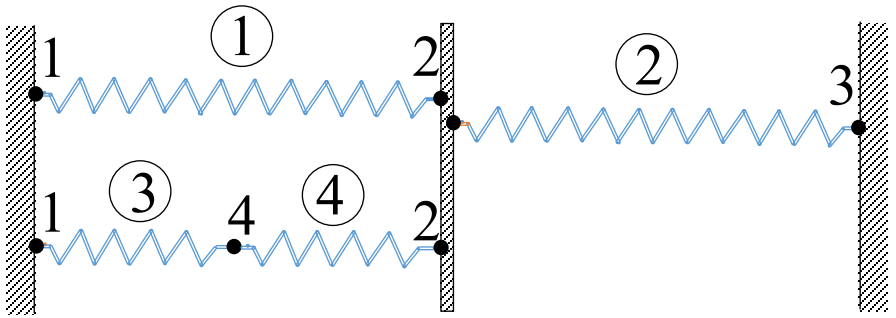
$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{bmatrix} [2] & [3] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$



$$\mathbf{K} = \begin{bmatrix} 3k & -3k & 0 & 0 \\ -3k & 3k & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \\ 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{bmatrix} [2] & [3] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

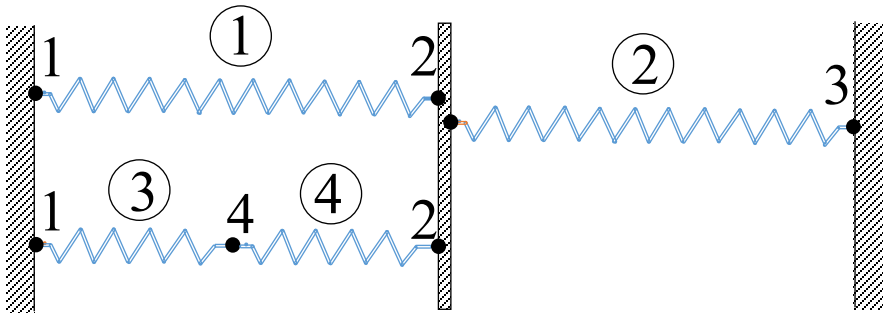


$$\mathbf{K} = \begin{bmatrix} 3k & -3k & 0 & 0 \\ -3k & 3k+k & -k & 0 \\ 0 & -k & k & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \\ 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{bmatrix} [2] & [3] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(3)} = \begin{matrix} \\ [1] \\ [4] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [4] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

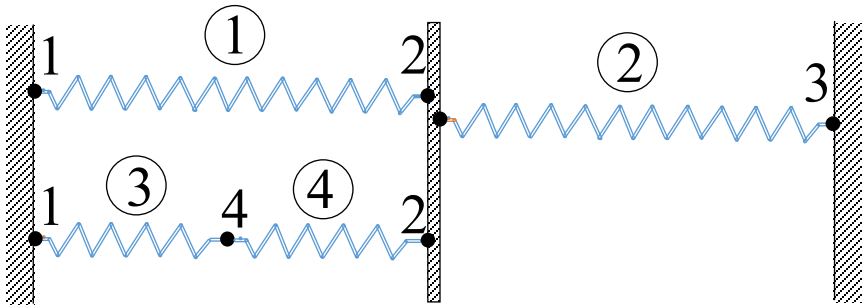


$$\mathbf{K} = \begin{bmatrix} 3k & -3k & 0 & 0 \\ -3k & 3k+k & -k & 0 \\ 0 & -k & k & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \\ 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{bmatrix} [2] & [3] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(3)} = \begin{matrix} \\ [1] \\ [4] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [4] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$



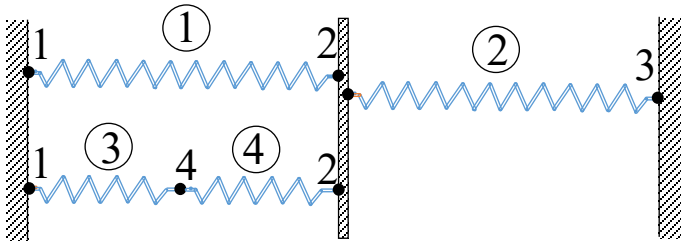
$$\mathbf{K} = \begin{bmatrix} 3k+k & -3k & 0 & -k \\ -3k & 3k+k & -k & 0 \\ 0 & -k & k & 0 \\ -k & 0 & 0 & k \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{matrix} [1] & [2] \end{matrix} \\ \begin{bmatrix} 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{matrix} [2] & [3] \end{matrix} \\ \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(3)} = \begin{matrix} \\ [1] \\ [4] \end{matrix} \begin{matrix} \begin{matrix} [1] & [4] \end{matrix} \\ \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(4)} = \begin{matrix} \\ [4] \\ [2] \end{matrix} \begin{matrix} \begin{matrix} [4] & [2] \end{matrix} \\ \begin{bmatrix} 2k & -2k \\ -2k & 2k \end{bmatrix} \end{matrix}$$



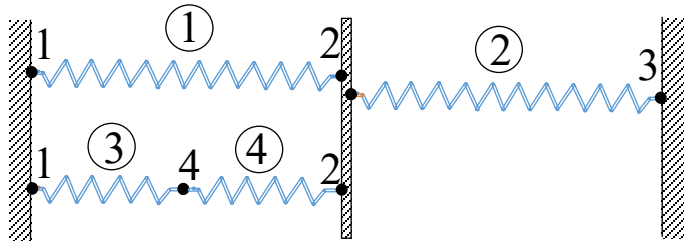
$$\mathbf{K} = \begin{bmatrix} 3k+k & -3k & 0 & -k \\ -3k & 3k+k & -k & 0 \\ 0 & -k & k & 0 \\ -k & 0 & 0 & k \end{bmatrix}$$

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \\ 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{bmatrix} [2] & [3] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(3)} = \begin{matrix} \\ [1] \\ [4] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [4] \\ k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(4)} = \begin{matrix} \\ [4] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [4] & [2] \\ 2k & -2k \\ -2k & 2k \end{bmatrix} \end{matrix}$$



$$\mathbf{K} = \begin{bmatrix} 3k+k & -3k & 0 & -k \\ -3k & 3k+k+2k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & k+2k \end{bmatrix}$$

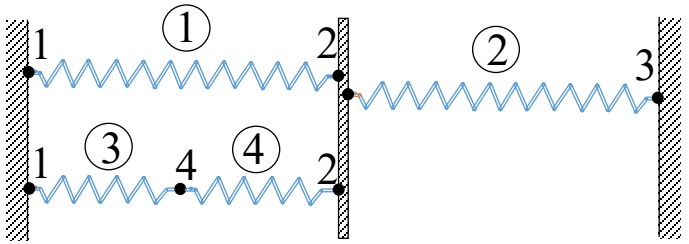
b2. Assembling element stiffness matrices

$$\mathbf{K}^{(1)} = \begin{matrix} \\ [1] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [2] \end{bmatrix} \\ \begin{bmatrix} 3k & -3k \\ -3k & 3k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{matrix} \\ [2] \\ [3] \end{matrix} \begin{matrix} \begin{bmatrix} [2] & [3] \end{bmatrix} \\ \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(3)} = \begin{matrix} \\ [1] \\ [4] \end{matrix} \begin{matrix} \begin{bmatrix} [1] & [4] \end{bmatrix} \\ \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \end{matrix}$$

$$\mathbf{K}^{(4)} = \begin{matrix} \\ [4] \\ [2] \end{matrix} \begin{matrix} \begin{bmatrix} [4] & [2] \end{bmatrix} \\ \begin{bmatrix} 2k & -2k \\ -2k & 2k \end{bmatrix} \end{matrix}$$



$$\mathbf{K} = \begin{bmatrix} 4k & -3k & 0 & -k \\ -3k & 6k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & 3k \end{bmatrix}$$

$$\begin{bmatrix} 4k & -3k & 0 & -k \\ -3k & 6k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & 3k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ \bar{u}_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} r_1 \\ 0 \\ r_3 \\ 0 \end{bmatrix}$$

c2. Partitioning the system

$$\mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L = \mathbf{f}_R^{ext} + \mathbf{r}_R$$

$$\mathbf{K}_{LR}\mathbf{d}_R + \mathbf{K}_{LL}\mathbf{d}_L = \mathbf{f}_L^{ext}$$

$$\begin{bmatrix} 4k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_3 \end{bmatrix}$$

$$\begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} 6k & -2k \\ -2k & 3k \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4k & -3k & 0 & -k \\ -3k & 6k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & 3k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ \bar{u}_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} r_1 \\ 0 \\ r_3 \\ 0 \end{bmatrix}$$

c2. Partitioning the system

$$\mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L = \mathbf{f}_R^{ext} + \mathbf{r}_R$$

$$\mathbf{K}_{LR}\mathbf{d}_R + \mathbf{K}_{LL}\mathbf{d}_L = \mathbf{f}_L^{ext}$$

$$\begin{bmatrix} 4k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_3 \end{bmatrix}$$

$$\begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} 6k & -2k \\ -2k & 3k \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4k & -3k & 0 & -k \\ -3k & 6k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & 3k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ \bar{u}_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} r_1 \\ 0 \\ r_3 \\ 0 \end{bmatrix}$$

c2. Partitioning the system

$$\mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L = \mathbf{f}_R^{ext} + \mathbf{r}_R$$

$$\mathbf{K}_{LR}\mathbf{d}_R + \mathbf{K}_{LL}\mathbf{d}_L = \mathbf{f}_L^{ext}$$

$$\begin{bmatrix} 4k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_3 \end{bmatrix}$$

$$\begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} 6k & -2k \\ -2k & 3k \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4k & -3k & 0 & -k \\ -3k & 6k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & 3k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ \bar{u}_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} r_1 \\ 0 \\ r_3 \\ 0 \end{bmatrix}$$

c2. Partitioning the system

$$\mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L = \mathbf{f}_R^{ext} + \mathbf{r}_R$$

$$\mathbf{K}_{LR}\mathbf{d}_R + \mathbf{K}_{LL}\mathbf{d}_L = \mathbf{f}_L^{ext}$$

$$\begin{bmatrix} 4k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_3 \end{bmatrix}$$

$$\begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} 6k & -2k \\ -2k & 3k \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4k & -3k & 0 & -k \\ -3k & 6k & -k & -2k \\ 0 & -k & k & 0 \\ -k & -2k & 0 & 3k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ \bar{u}_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} r_1 \\ 0 \\ r_3 \\ 0 \end{bmatrix}$$

c2. Partitioning the system

$$\mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L = \mathbf{f}_R^{ext} + \mathbf{r}_R$$

$$\mathbf{K}_{LR}\mathbf{d}_R + \mathbf{K}_{LL}\mathbf{d}_L = \mathbf{f}_L^{ext}$$

$$\begin{bmatrix} 4k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_3 \end{bmatrix}$$

$$\begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_3 \end{bmatrix} + \begin{bmatrix} 6k & -2k \\ -2k & 3k \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \end{bmatrix}$$

$$\mathbf{d}_L = \mathbf{K}_{LL}^{-1}\left(\mathbf{f}_L^{ext} - \mathbf{K}_{LR}\mathbf{d}_R\right)$$

$$\begin{bmatrix} u_2 \\ u_4 \end{bmatrix} = \begin{bmatrix} 6k & -2k \\ -2k & 2k \end{bmatrix}^{-1} \left\{ \begin{bmatrix} 50 \\ 0 \end{bmatrix} - \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} \overline{u}_1 \\ \overline{u}_3 \end{bmatrix} \right\}$$

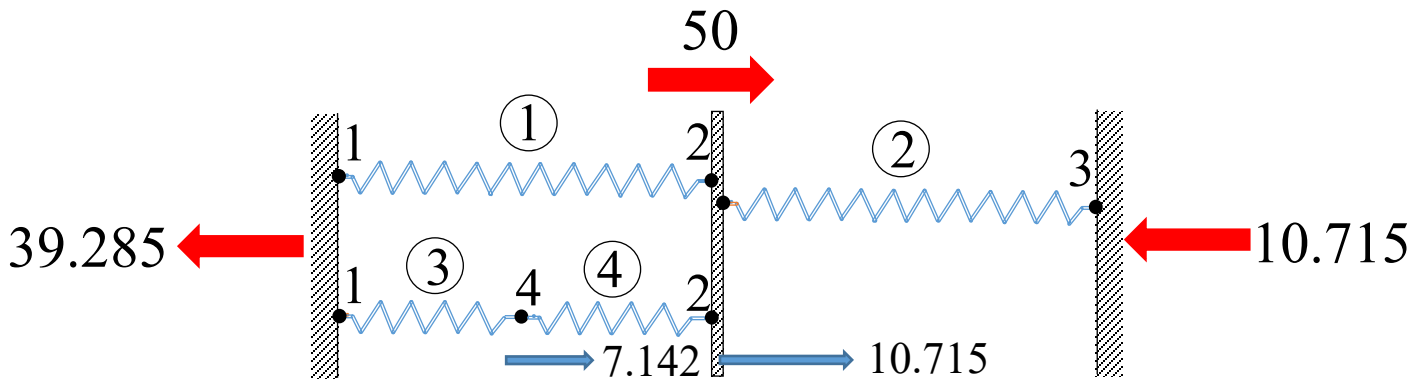
d. Compute the reaction forces

$$\mathbf{r}_R = \mathbf{K}_{RR}\mathbf{d}_R + \mathbf{K}_{RL}\mathbf{d}_L - \mathbf{f}_R^{ext}$$

$$\begin{bmatrix} r_1 \\ r_3 \end{bmatrix} = \begin{bmatrix} 4k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} \overline{u}_1 \\ \overline{u}_3 \end{bmatrix} + \begin{bmatrix} -3k & -k \\ -k & 0 \end{bmatrix} \begin{bmatrix} u_2 \\ u_4 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

For $k=1$

$$\mathbf{d}=\begin{bmatrix}\overline{u}_1\\ u_2\\ \overline{u}_3\\ u_4\end{bmatrix}=\begin{bmatrix}0\\ 10.715\\ 0\\ 7.142\end{bmatrix} \qquad \mathbf{r}=\begin{bmatrix}\overline{r}_1\\ r_2\\ \overline{r}_3\\ r_4\end{bmatrix}=\begin{bmatrix}-39.285\\ 0\\ -10.715\\ 0\end{bmatrix}$$

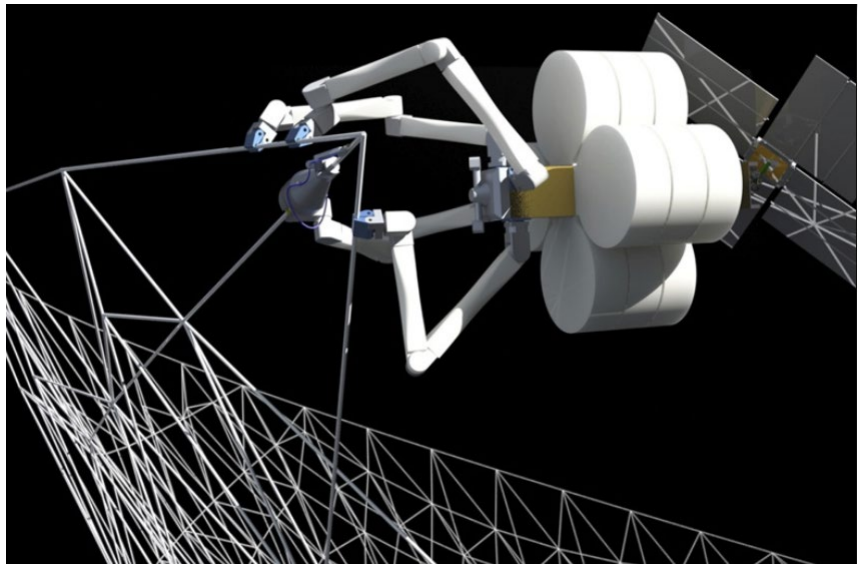


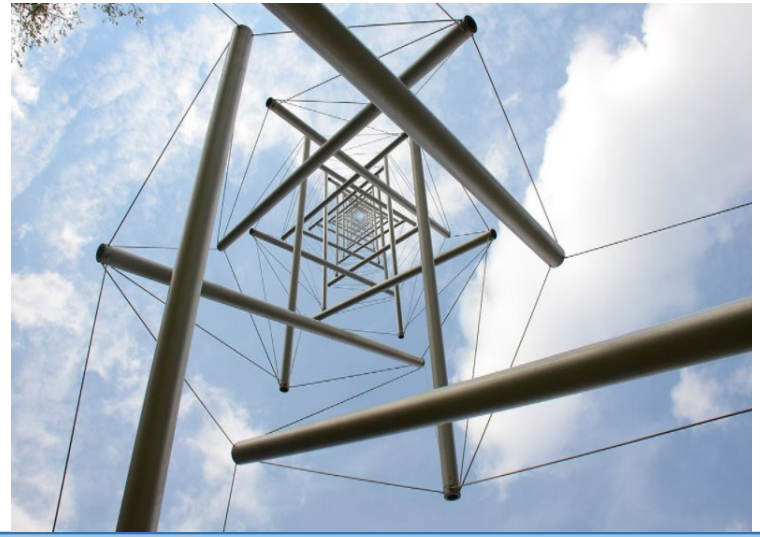
3D TRUSS STRUCTURES

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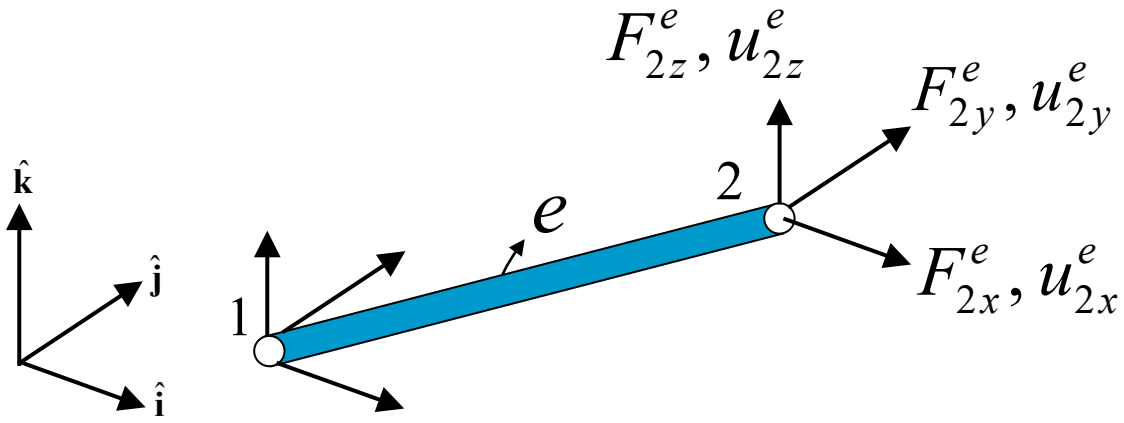




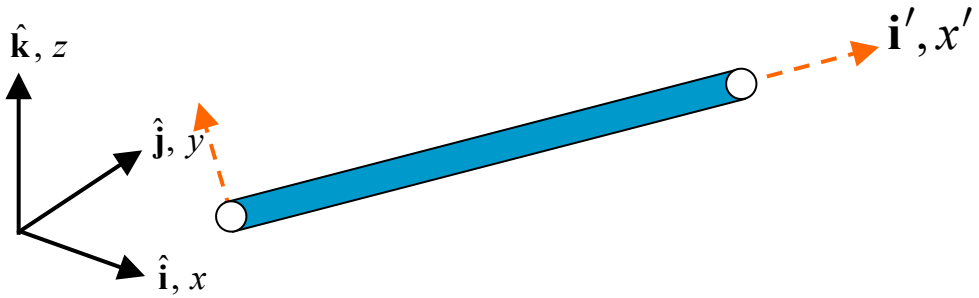




Bar element (3-D case)



$$F^e = \begin{bmatrix} F^e_{1x} \\ F^e_{1y} \\ F^e_{1z} \\ F^e_{2x} \\ F^e_{2y} \\ F^e_{2z} \end{bmatrix} \qquad d^e = \begin{bmatrix} u^e_{1x} \\ u^e_{1y} \\ u^e_{1z} \\ u^e_{2x} \\ u^e_{2y} \\ u^e_{2z} \end{bmatrix}$$

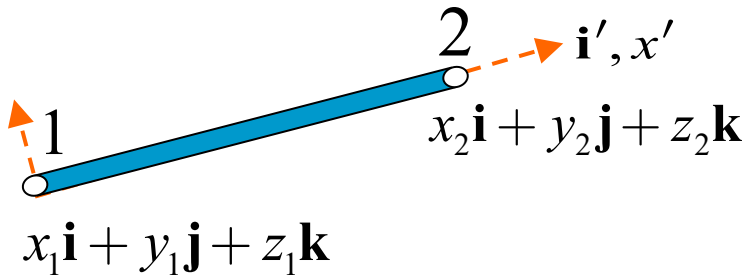


Relationship between forces and displacements in the local coordinate system

$$\begin{bmatrix} F'^e_{1x} \\ F'^e_{2x} \end{bmatrix} = \underbrace{\begin{bmatrix} k^e & -k^e \\ -k^e & k^e \end{bmatrix}}_{\mathbf{K}'^e} \begin{bmatrix} u'^e_{1x} \\ u'^e_{2x} \end{bmatrix}$$

The new stiffness matrix

- Construction of the rotation matrix $\mathbf{R}^e$



1) Construction of the local coordinate system

$$\mathbf{i}' = \frac{1}{l^e}\left((x_2-x_1)\mathbf{i}+(y_2-y_1)\mathbf{j}+(z_2-z_1)\mathbf{k}\right)$$

2) Nodal displacement in the two coordinate systems (local and global)

$$\underbrace{u'^e_{1x}\,\mathbf{i}'+u'^e_{1y}\,\mathbf{j}'+u'^e_{1z}\,\mathbf{k}'}_{\mathbf{u}_1} = \underbrace{u^e_{1x}\,\mathbf{i}+u^e_{1y}\,\mathbf{j}+u^e_{1z}\,\mathbf{k}}_{\mathbf{u}_1}$$

3) Determination of the first component of the local displacement

$$u'^e_{1x} = u^e_{1x}\,\mathbf{i}\cdot\mathbf{i}'+u^e_{1y}\,\mathbf{j}\cdot\mathbf{i}'+u^e_{1z}\,\mathbf{k}\cdot\mathbf{i}'$$

$$u'^e_{1x} = \frac{1}{l^e}\left[u^e_{1x}(x_2-x_1)+u^e_{1y}(y_2-y_1)+u^e_{1z}(z_2-z_1)\right]$$

4) Application of similar concepts to the other node

$$u'^{e}_{2x} = \frac{1}{l^e}\left[u^e_{2x}(x_2 - x_1) + u^e_{2y}(y_2 - y_1) + u^e_{2z}(z_2 - z_1)\right]$$

5) New relation between $\mathbf{d}'^{e}$ and $\mathbf{d}^e$

$$\begin{bmatrix} u'^{e}_{1x} \\ u'^{e}_{2x} \end{bmatrix} = \underbrace{\frac{1}{l^e}\begin{bmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \end{bmatrix}}_{\mathbf{R}^e} \underbrace{\begin{bmatrix} u^e_{1x} \\ u^e_{1y} \\ u^e_{1z} \\ u^e_{2x} \\ u^e_{2y} \\ u^e_{2z} \end{bmatrix}}_{\mathbf{d}^e}$$

6) Local internal force and global displacement

$$\begin{bmatrix} F'^{e}_{1x} \\ F'^{e}_{2x} \end{bmatrix} = \begin{bmatrix} k^e & -k^e \\ -k^e & k^e \end{bmatrix} \begin{bmatrix} u'^{e}_{1x} \\ u'^{e}_{2x} \end{bmatrix} = \mathbf{K}'^{e} \begin{bmatrix} u'^{e}_{1x} \\ u'^{e}_{2x} \end{bmatrix} = \mathbf{K}'^{e}\mathbf{R}^e\mathbf{d}^e$$

7) Global and local internal force

$$\begin{bmatrix} F'^{e}_{1x} \\ F'^{e}_{2x} \end{bmatrix} = \underbrace{\frac{1}{l^e}\begin{bmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_2-x_1 & y_2-y_1 & z_2-z_1 \end{bmatrix}}_{\mathbf{R}^e} \underbrace{\begin{bmatrix} F^e_{1x} \\ F^e_{1y} \\ F^e_{1z} \\ F^e_{2x} \\ F^e_{2y} \\ F^e_{2z} \end{bmatrix}}_{\mathbf{F}^e}$$

8) New global force-displacement relationship (details of the validity of the following expressions can be studied in the appendix , at the end of this section)

$$\mathbf{F}^e = \mathbf{R}^{eT}\mathbf{F}'^e = \mathbf{R}^{eT}\mathbf{K}'^e\mathbf{d}'^e = \underbrace{\mathbf{R}^{eT}\mathbf{K}'^e\mathbf{R}^e}_{\mathbf{K}^e}\mathbf{d}^e$$

9) Global stiffness matrix for 3D

$$\mathbf{K}^e = \underbrace{\mathbf{R}^{eT}}_{6\text{x}2}\underbrace{\mathbf{K}'^e}_{2\text{x}2}\underbrace{\mathbf{R}^e}_{2\text{x}6}$$